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Intelligent control of
robot arm based on LLM

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Abstract

In complex stacked and occluded environments, robotic grasping systems often suffer from incomplete perception, environmental uncertainty, and the difficulty of constructing accurate system models. Traditional approaches that rely solely on geometric perception and fixed control strategies are therefore unable to guarantee stable and reliable task execution. From the perspective of intelligent control theory, this class of problems can be fundamentally formulated as a partially observable hybrid control problem, in which high-level discrete strategy selection must be tightly coupled with low-level continuous motion control under physical constraints.

To address this challenge, this paper proposes a hierarchical closed-loop intelligent control framework for occluded robotic grasping. The proposed framework adopts a five-layer architecture consisting of perception, semantic reasoning, world modeling, decision and task planning, and motion control. By integrating conventional geometric perception with high-level semantic reasoning, visual outputs are no longer treated as deterministic states but as partial observations. A semantic–geometric aligned world modeling mechanism is further introduced to construct a decision-oriented belief state, effectively alleviating world model mismatch caused by occlusion and illumination variations.

Based on this belief state, a belief-driven hybrid decision controller is designed, which abstracts complex continuous perceptual information into a low-dimensional discrete cognitive state. Through a mode-switching strategy, the system adaptively selects among three operational modes: abort, direct grasping, and occluder removal, thereby achieving stable closed-loop control in complex environments. The high-level semantic decisions are further grounded into physically executable actions via a semantic-to-motion mapping mechanism combined with inverse kinematics and feasibility verification, forming a complete cognition–decision–execution–feedback intelligent control loop.

Extensive qualitative and quantitative experiments demonstrate that, under weak perception conditions such as severe occlusion and degraded illumination, the

proposed method significantly outperforms traditional geometry-only strategies in terms of target existence determination, occlusion reasoning, and occluder classification, while maintaining stable and consistent closed-loop decision behavior. These results verify the effectiveness of the proposed framework for achieving robust intelligent control under partial observability.

Keywords: Intelligent control; Robotic manipulator; Partial observability; Hybrid control; Occluded grasping; Belief state; Semantic–geometric alignment; Hierarchical control architecture; Robotic manipulation

要旨 (Abstract)

複雑な積み重ねおよび遮蔽環境において、ロボットマニピュレータの把持システムは、知覚の不完備性、環境の不確実性、ならびに精確なシステムモデル構築の困難さといった課題に直面する。幾何情報のみに基づく従来手法や固定制御戦略では、安定かつ信頼性の高いタスク実行を保証することは困難である。

知能制御理論の観点から見ると、本問題は本質的に部分可観測ハイブリッド制御問題として定式化でき、高レベルの離散的戦略選択と、物理拘束下における低レベルの連続運動制御とを密接に結合して扱う必要がある。

本研究では、遮蔽環境下のロボット把持を対象とした分層型閉ループ知能制御フレームワークを提案する。本フレームワークは、知覚、意味推論、世界モデル構築、意思決定・タスク計画、運動制御の五層構造から構成される。従来の幾何知覚に高レベルの意味推論を統合することで、視覚出力を決定的な状態としてではなく部分観測として扱い、さらに意味-幾何整合型世界モデルを導入することで、遮蔽や照明変動による世界モデルの不整合を効果的に緩和する。

構築された信念状態 (belief state) に基づき、信念駆動型ハイブリッド意思決定制御器を設計し、連続的な知覚情報を低次元の離散認知状態へと抽象化する。これにより、中止、直接把持、遮蔽物除去の三つの制御モード間を自律的に切り替え、複雑環境下における安定な閉ループ制御を実現する。高レベルの意味決定は、逆運動学および可行性検証と組み合わせた意味から運動へのマッピングを通じて物理的に実行可能な動作へと変換され、認知-意思決定-実行-フィードバックからなる知能制御ループを構成する。

多数の定性および定量実験の結果、強い遮蔽や低照度といった弱知覚条件下においても、提案手法は、目標存在判定、遮蔽推論、遮蔽物分類の各指標において、幾何情報のみに基づく従来手法を大きく上回り、安定かつ一貫した閉ループ意思決定挙動を維持できることが確認された。これらの結果は、本フレームワークが部分可観測環境下におけるロバストな知能制御を実現する有効な手法であることを示している。

キーワード: 知能制御; ロボットマニピュレータ; 部分可観測; ハイブリッド制御; 遮蔽把持; 信念状態; 意味-幾何整合; 分層制御アーキテクチャ; ロボット操作

Table of Contents

1 INTRODUCTION	3
1.1 Intelligent Control Problem of Robot Arms	3
1.2 Control Limitations of Existing Approaches	5
1.3 System Framework and Research Objectives	5
1.4 Illustration of a Typical Occluded Scenario under Weak Perception	6
1.5 Main Contributions of This Work	8
2 Related Work	9
2.1 Vision as Partial Observation under Occlusion	9
2.2 World Modeling under Perceptual Uncertainty	10
2.3 Grounded Semantic–Geometric Representation	11
2.4 Hybrid Decision-Making and Motion Planning under Occlusion	12
2.5 Semantic-to-Motion Control	12
2.6 Summary of Related Work and Positioning	12
3 Approach	14
3.1 Perception Layer	18
3.2 World Modeling Layer	19
3.3 Grounded World Modeling Layer	22
3.4 Decision and Task Planning Layer	25
3.4.1 Target Selection	27
3.4.1.1 Target Definition and Task Semantics	27
3.4.1.2 Decision Function Logic	27
3.4.1.3 Target Instance Selection	28
3.4.2 Target Occlusion Determination	29
3.4.2.1 VLM-based Occlusion Reasoning for Target Verification	29
3.4.2.2 Structured Occlusion Reasoning Result Set $\mathcal{R}$	30
3.4.2.3 Definition of the Occlusion Decision Function	30
3.4.3 High-Level Decision-Making and Target Anchoring	31
3.4.3.1 Decision States and Discrete Control Modes	31
3.4.3.2 Execution Object Anchoring	32
3.4.3.3 Closed-Loop Grasping Control under Occlusion	33
3.5 Semantic-to-Motion Control Layer	34
3.5.1 Binding Semantic Objects to Geometric Targets	34
3.5.2 Geometric Coordinate Extraction and Spatial Projection	35
3.5.2.1 Pixel Coordinate Decoding and Distortion Correction	35
3.5.2.2 Ray Projection from Image to Camera Coordinate System	36
3.5.2.3 Hand–Eye Calibration and Coordinate Transformation	37
3.5.2.4 Depth Recovery from a Planar Constraint	40
3.5.3 Five-DOF Inverse Kinematics and Feasibility Check	41
3.5.3.1 Base Joint Angle θ_1	43
3.5.3.2 Shoulder Joint θ_2	43
3.5.3.3 Elbow Joint θ_3	44
3.5.3.4 Wrist Pitch θ_4	45
3.5.3.5 Gripper Swing θ_5	45

3.5.3.6 Closed-Form IK Summary and Feasibility Verification	45
3.5.4 Embedded Execution Mapping	46
3.6 Summary of the Proposed Control Framework	47
4 Experimental and Results Analysis	49
4.1 Experimental Platform and System Configuration	49
4.1.1 Actuator Design	50
4.1.2 Experimental Platform Computer System	50
4.2 Experiment I: Scenarios under Occluded and Non-occluded Conditions	51
4.2.1 Scene Categorization and Experimental Motivation	52
4.2.2 Experimental Results and Analysis under Occluded and Non-occluded Scenes	53
4.2.2.1 Occluded Scene · Normal Case (Case A)	53
4.2.2.2 Occluded Scene · Effective Case (Case B)	54
4.2.2.3 Non-occluded Scene · Normal Case (Case C)	55
4.2.2.4 Non-occluded Scene · Effective Case (Case D)	56
4.2.3 Performance Summary under Occluded Conditions	57
4.3 Experiment II: Perceptual Ambiguity under Illumination in Occluded and Adjacent Scenarios ..	58
4.3.1 Experimental Motivation and Definition of Influencing Factors	58
4.3.2 Experimental Data and Scene Construction	59
4.3.2.1 Scene Types	59
4.3.2.2 Illumination Control and Calibration	60
4.3.2.3 Data Collection and Annotation	60
4.3.3 Comparison Methods	61
4.3.4 Definition of Evaluation Metrics	62
4.3.5 Experimental Results and Analysis	63
4.3.5.1 Partially Occluded Scenes	63
4.3.5.2 Adjacent Non-Occluded Scenes	65
4.3.5.3 Overall Observation and Methodological Difference	66
4.4 Experiment III: Simulation-to-Real Transfer in End-to-End Semantic Robotic Grasping	66
4.4.1 Experiment Objective	66
4.4.2 Experiment Setup and Procedure	67
4.4.3 System-Level End-to-End Execution Case Study	69
4.4.4 System-Level Evaluation Metrics and Failure Attribution	72
4.4.5 Experiment Data and Analysis	73
5 Conclusions	75
References	78
Appendix A	80
Appendix B	82
Appendix C	84
Acknowledgements	88

1 INTRODUCTION

1.1 Intelligent Control Problem of Robot Arms

In control theory, intelligent control is commonly used to describe a class of systems in which the controlled plant and its environment cannot be accurately modeled using conventional differential or difference equations, and where significant uncertainty and incomplete observations are present. Under such conditions, relying solely on fixed feedback control or traditional adaptive control is often insufficient to meet task requirements. Instead, the control system must incorporate task planning, reasoning, structural reconfiguration, learning mechanisms, and hybrid control strategies, establishing a cooperative relationship between continuous dynamic control and discrete decision-making processes, thereby enabling autonomous regulation and stable operation of complex systems [1].

In the field of robotics, intelligent control further emphasizes a hierarchical cooperative architecture between high-level cognitive decision-making and low-level physical control. Through multi-source perception and high-level reasoning, the system constructs a world model and belief state of the environment, performs discrete strategy selection and control-mode switching at the high level, executes continuous motion control at the low level, and continuously corrects its understanding of the environment and its own state through closed-loop feedback. This “cognition–control” synergy allows robots to sustain a closed-loop cycle of understanding–decision–execution–correction in dynamic, uncertain, and perception-limited environments [2].

Under the above theoretical framework, this study focuses on a representative robotic intelligent control problem: grasping decision and control of a robot arm under weak perception conditions.

Weak perception refers to situations in which the robot cannot reliably obtain a complete and accurate representation of the scene due to factors such as occlusion,

close object proximity, illumination variations, or limited viewpoints. Under such conditions, even when the target object is not geometrically occluded, it may still be misclassified as “occluded” in the two-dimensional visual projection simply because it is spatially adjacent to surrounding objects, thereby introducing systematic discrepancies between the world model and the real scene.

In real-world applications—such as warehouse picking, logistics sorting, and food service—target objects are often embedded in cluttered environments with stacking, occlusion, and dense spatial proximity. The success of a grasp depends not only on the geometric shape and pose of the target itself, but also critically on its spatial relationships with neighboring objects, reachability constraints, and potential physical interferences. Consequently, the system dynamics exhibit strong nonlinearity, strong coupling, and uncertainty, making it difficult to construct accurate analytical models using traditional differential or difference equations.

From a control perspective, this problem can be formulated as a partially observable hybrid control problem [3]. The system must continuously regulate joint angles and end-effector trajectories in a continuous space, while also making high-level discrete decisions between operational modes such as directly grasping the target or first removing obstructing objects [4][5], which leads to a characteristic coupling between discrete and continuous dynamics.

Under weak perception, the true occlusion relationships and graspable regions cannot be fully observed, and the system can only construct an uncertain world model based on incomplete sensory information. The controller is therefore no longer faced with a single deterministic state, but must reason over multiple possible states and select appropriate strategies accordingly. This problem thus goes beyond traditional control paradigms that rely on precise models and exact state feedback; its core lies in cognitive reasoning and strategic decision-making for unmodeled and partially observable systems. Therefore, robot arm grasping under weak perception constitutes example of an intelligent control problem.

1.2 Control Limitations of Existing Approaches

Most existing robot arm grasping systems first employ object detection and instance segmentation methods [6][7] to perceive the geometric information of objects from images, and then perform reachability analysis and grasp planning based on the perceived results. While such approaches work well in scenes where objects are clearly visible and occlusion is limited, they often exhibit unstable performance in realistic occluded environments.

In our preliminary experiments, we observe that instance segmentation masks frequently suffer from errors when object boundaries overlap or when illumination conditions deteriorate. These errors typically manifest as discontinuities along object boundaries or as regions being absorbed by the contours of neighboring objects [8]. For example, when a cup is partially occluded by a fork or a spoon, its contour is often not completely segmented. Under such conditions, the geometric world model constructed by the system deviates significantly from the true physical state.

From a control-theoretic perspective, this constitutes a typical world model mismatch problem [9]. In such cases, the system performs decision-making based on an inaccurate perceptual representation. Although reachability analysis and collision checking may be geometrically correct in principle, their inputs are already erroneous or incomplete. This mismatch between the world model and the physical environment can cause the control system to incorrectly assess grasp feasibility, leading to collisions or grasp failures. The root cause of this problem lies not in the trajectory planning algorithms themselves, but in the insufficient reliability of high-level state representations.

1.3 System Framework and Research Objectives

To address the aforementioned challenges, this study proposes a hierarchical robot arm control framework for occluded environments. The core idea is to augment

conventional geometry-based perception with semantic information during world model construction, enabling the system to obtain relatively stable scene understanding even when the target object is partially occluded and observations are incomplete.

In the proposed system, the perception layer first extracts geometric information of the target object and surrounding objects from images, such as object categories, bounding boxes, and relative spatial relationships. Due to occlusion and environmental factors, these perceptual results are often incomplete and therefore require further modeling and interpretation. By integrating geometric information with semantic cues, the system infers both the existence of the target and its occlusion state, thereby forming a belief estimate of the current scene [10].

The proposed Decision and Motion Planning layer does not rely directly on raw detection results. Instead, it dynamically selects among different operational modes, including abandoning the grasp, directly grasping the target, or first removing occluding objects. The primary focus of this research is not to improve the accuracy of individual perception algorithms, but rather to enhance system-level robustness through architectural design, enabling the robot arm to maintain stable and reliable grasping decisions and execution under incomplete perceptual information.

1.4 Illustration of a Typical Occluded Scenario under Weak Perception

As illustrated in Fig.1-1, our preliminary experiments show that instance segmentation results become notably unstable when the target cup is occluded by objects such as spoons or forks. In particular, when critical graspable regions—such as the rim of the cup—are partially occluded, these regions are often absorbed by the contours of the occluding objects, causing the segmentation mask of the cup to fragment.

Under such conditions, the system finds it difficult to reliably determine whether

the target object truly exists in the current scene, which directly affects subsequent decision-making and grasp execution.



Fig. 1-1. Robot Arm Grasping in a Real-World Physical Environment

To address perceptual incompleteness caused by occlusion, this study does not treat a single visual output as the true state of the physical world. Instead, each observation is regarded as a partial observation of the scene and is incorporated into a semantic – geometric world model, within which a high-level reasoning module estimates both target existence and occlusion relationships. Through this process, the system is able to determine whether the target object exists even under missing or incomplete information.

Once the existence of the target object is confirmed, the system further evaluates whether it is occluded by other objects. If the target is determined to be occluded, the system prioritizes generating grasp-and-removal actions for the occluding objects (e.g., spoons or forks), rather than directly attempting to grasp the cup. After executing an occluder removal action, the system performs visual perception again and feeds the new observation back into the world model to update the scene state. Through this closed-loop process of observation – reasoning – action – re-observation, the system progressively reduces uncertainty in the target region until the target object

becomes stable and clearly identifiable.

Within this iterative loop, grasping behavior is no longer treated as a one-shot decision. Instead, it gradually converges toward effective control of the target object based on continuously updated scene states. This closed-loop control mechanism, centered on semantic compensation and occlusion removal, enables the system to stably execute a sequence of actions from occluder removal to final target grasping, even in complex occluded environments.

1.5 Main Contributions of This Work

The main contributions of this study are summarized as follows:

- (1) A hierarchical decision-making and motion control framework for robot arm in occluded environments is proposed, enabling the system to stably switch among different operational modes under partially observable conditions.
- (2) A semantic – geometric aligned world model and belief estimation mechanism is designed to map incomplete visual observations into high-level state representations suitable for control and planning.
- (3) Extensive experiments on a real robot arm platform are conducted under complex stacked and occluded scenarios, validating the stability and effectiveness of the proposed control architecture under incomplete perception.

2 Related Work

To achieve stable and reliable grasping in complex occluded environments, robotic systems must not only possess strong perception capabilities, but also perform rational decision-making and motion control under conditions of incomplete observation and state uncertainty. From a control-theoretic perspective, such problems are commonly formulated as Partially Observable Markov Decision Processes (POMDPs) or belief-space control problems [11][12], whose core lies in realizing stable behavior in uncertain environments through a closed-loop structure of observation, state estimation, decision, and control. In recent years, belief-space planning and hybrid control frameworks [13] have become important theoretical foundations for addressing such challenges. In occluded grasping tasks, robots must resolve three tightly coupled issues:

- (1) how to treat visual outputs as observations rather than deterministic states under incomplete perception;
- (2) how to construct a world model that supports reasoning and decision-making (state/belief representation);
- (3) how to perform hybrid discrete mode selection and continuous motion control under state uncertainty.

Following this closed-loop control pipeline, this chapter systematically reviews related work from the perspectives of visual observation, world modeling, semantic – geometric alignment, hybrid decision-making and motion planning under occlusion, and semantic-to-motion control, and analyzes the limitations of existing approaches in complex occluded and weak-perception conditions.

2.1 Vision as Partial Observation under Occlusion

In robot arm grasping systems, vision is the primary information source for high-level decision-making and control. However, from a control-theoretic

perspective, visual outputs can only be regarded as partial observations of the true world state, rather than deterministic states that can be directly used for planning [11].

In recent years, object detection and instance segmentation networks have achieved remarkable accuracy in complex scenes. For example, DetectoRS proposed by Qiao et al. [14] employs recursive feature pyramids and switchable atrous convolution, achieving over 55% detection AP and 48% segmentation AP on the COCO dataset. BCNet proposed by Ke et al. [15] explicitly models occluding and occluded layers, significantly improving segmentation accuracy in overlapping-object scenarios. Lightweight models such as SparseInst by Cheng et al. [16] further achieve high instance segmentation performance while maintaining real-time efficiency.

Despite representing the performance ceiling of 2D perception models, these methods still struggle in multi-object stacking and heavy occlusion scenes, where targets often expose only partial contours. As a result, bounding boxes and segmentation masks are frequently fragmented or misclassified. This implies that visual outputs are inherently local, noisy, and incomplete observations. When such outputs are directly treated as deterministic world states and fed into downstream planning and control modules, severe state mismatches may occur, leading to unstable grasping decisions.

2.2 World Modeling under Perceptual Uncertainty

To transform visual observations into state representations usable for planning and decision-making, robotic systems typically construct a world model that describes the spatial distribution, attributes, and relationships of objects in the scene. From a control perspective, the world model serves the role of state estimation or belief modeling [12].

Early works employed scene graphs to represent objects and their spatial relationships [16]. With the development of RGB-D perception and 3D reconstruction techniques, world models have gradually evolved into 3D scene graphs. The 3D

Dynamic Scene Graph (DSG) proposed by Rosinol et al. [17] uses a hierarchical graph structure to unify objects, spatial regions, and dynamic entities, enabling the world model to support both low-level motion planning and high-level task reasoning. ConceptGraphs proposed by Gu et al. [18] fuse multi-view RGB-D observations into an open-vocabulary 3D scene graph and leverage Vision – Language Models to generate semantic labels. The Functional 3D Scene Graph proposed by Zhang et al. [19] further incorporates functional semantics, allowing the world model to encode object affordances.

However, these methods generally rely on reliable perception and geometric reconstruction. When occlusions are severe, segmentation fails, or depth noise is present, object positions and topological relationships are incorrectly estimated and propagated through the world model to the decision layer, causing the system to generate incorrect actions under false state assumptions.

2.3 Grounded Semantic–Geometric Representation

To bridge high-level task semantics with physical geometric entities, recent studies have focused on semantic–geometric alignment. Schwarz et al. [20] align RGB-D object detection results with semantic segmentation regions, establishing a consistent mapping between object identities and their physical occupancy. The PASG framework proposed by Zhu et al. [21] associates geometric primitives with high-level task semantics through semantic anchoring. The SoFAR framework proposed by Qi et al. [22] further introduces semantic orientations, aligning geometric poses with natural language functional descriptions.

From a control perspective, these approaches introduce high-level semantic state variables into the world model. However, they usually assume that object geometry is complete and reliable, and do not explicitly model occlusion, uncertainty, or reachability. When a target is only partially visible, such grounded representations struggle to maintain stable assumptions about the true world state.

2.4 Hybrid Decision-Making and Motion Planning under Occlusion

In stacked and occluded environments, manipulators often need to remove occluding objects before grasping the target, which requires discrete–continuous hybrid control decisions on top of the world model. Task and Motion Planning (TAMP) frameworks [23] realize this hybrid structure by jointly solving discrete task decisions and continuous motion constraints.

The push–grasping framework proposed by Dogar and Srinivasa [24] identifies occluding objects using swept volume analysis [25] and recursively generates removal actions. However, this method is highly dependent on geometric model accuracy. Under incomplete perception, occlusion relationships are often misjudged, leading to incorrect mode selection and task failure.

2.5 Semantic-to-Motion Control

Recently, LLM and VLM have been introduced into robotic control, enabling semantic information to directly modulate physical actions. Vision–Language–Action (VLA) models such as RT-2 [26] learn end-to-end mappings from semantic inputs to motor actions, but lack explicit world models and occlusion reasoning capabilities. VoxPoser [27] constructs a 3D value map to project semantic constraints into control objectives, enabling interpretable semantic-to-motion control, but still assumes reliable perception.

2.6 Summary of Related Work and Positioning

Current intelligent control approaches for weak-perception robotic manipulation can be broadly categorized into two classes:

(1) Geometry + TAMP methods: possess hybrid control structures, but strongly depend on perception accuracy;

(2) End-to-end VLA methods: exhibit semantic generalization, but lack explicit state representations and world models.

From an intelligent control perspective, the system proposed in this work treats perception as observation, the world model as state/belief estimation, and constructs a decision-ready belief state through semantic–geometric alignment. On this basis, it performs discrete mode switching and continuous motion control. The proposed framework can therefore be regarded as a model-driven hybrid intelligent control system operating under partial observability.

3 Approach

This study proposes a hierarchical closed-loop grasping control system for occluded environments. In the proposed system, the upper layer is responsible for constructing and maintaining a cognitive state under incomplete perception and performing discrete decision-making, while the lower layer executes verifiable motion control in continuous physical space. This layered design enables stable and reliable grasping behavior in complex occluded scenarios.

Structurally, the proposed framework shares similarities with recent hierarchical control architectures such as GR00T-N1 [28]. However, the key distinction lies in the introduction of explicit semantic reasoning for occlusion at the high level via a VLM. This design improves decision stability under occlusion while preserving the determinism and reliability of low-level motion control.

As illustrated in Fig. 3-1, the system is implemented as a complete closed-loop pipeline from visual perception and semantic reasoning to grasp execution. First, a YOLO-based module performs object detection and instance segmentation on RGB images captured by the camera, producing object categories, bounding boxes, and center coordinates, along with annotated images. The detected geometric measurements and annotated images are then packaged together with the task instruction and fed into a LLM. Guided by predefined prompts, the LLM performs occlusion-aware decision-making and outputs the pixel coordinates of the selected grasp target.

Subsequently, the system maps the selected pixel coordinates to physical world coordinates. An inverse kinematics (IK) module computes the corresponding joint angles, and the grasp feasibility is verified through simulation. If the grasp is deemed infeasible, the system aborts the execution. Otherwise, the robot arm performs the grasping action. The camera continues to capture scene images after execution, and the system proceeds to the next grasping cycle, thereby forming a closed-loop control process.

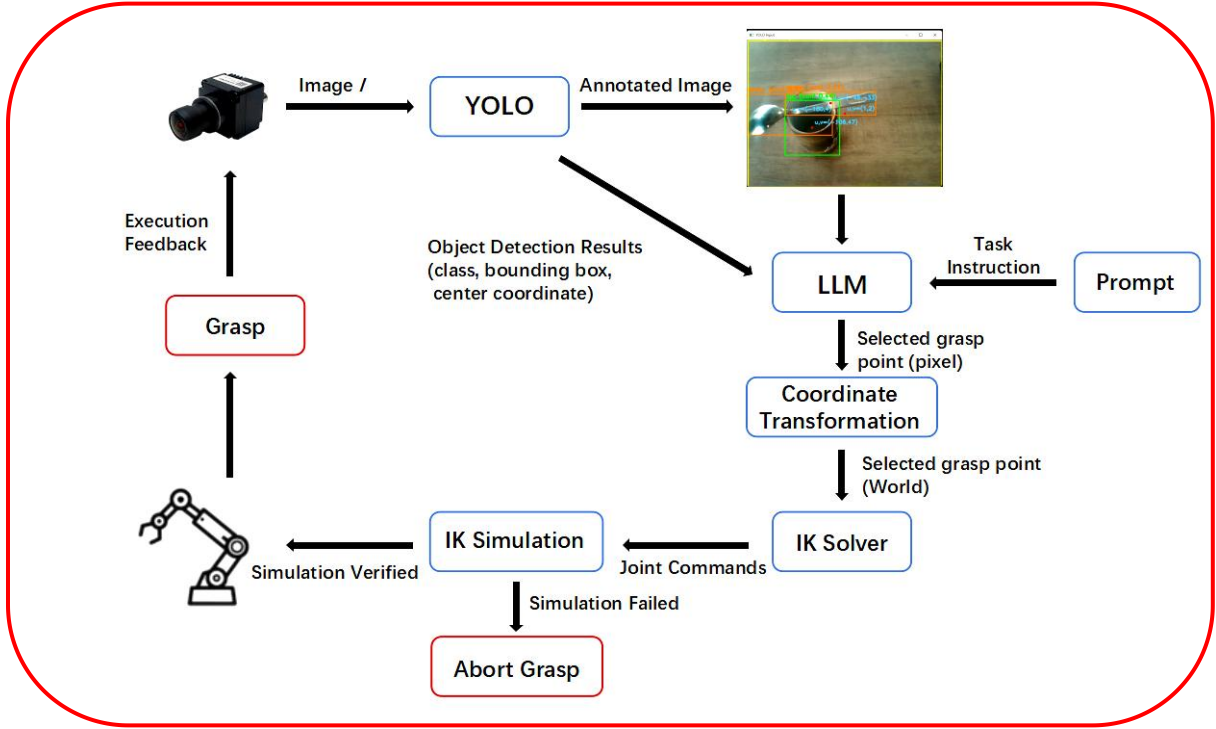


Fig. 3-1. Overall Workflow of the Robot Arm Grasping Decision System

As illustrated in Fig. 3-1, the proposed system is not a one-shot open-loop grasping pipeline, but a vision-based closed-loop grasping control system.

In each execution cycle, the robotic arm first returns to a predefined initial observation pose, where a fixed camera captures the current scene as the system observation. The acquired image is then processed by the decision-making module to generate the grasping decision and the target pose parameters.

After the grasping action is executed and execution feedback is obtained, the robotic arm returns again to the initial observation pose, re-acquires a new image, and enters the next perception–decision–execution loop. This process iterates continuously until the target object is successfully grasped or the system determines that the task has failed.

(1) Perception Layer

The Perception Layer is responsible for acquiring raw multimodal data from the physical environment, such as RGB-D images. Object detection and instance segmentation algorithms (e.g., YOLO) are applied to transform these raw observations into structured environmental descriptions, extracting object class

attributes, bounding boxes, and confidence scores. These outputs provide fundamental visual priors for subsequent layers.

(2) World Modeling Layer

Under incomplete perception, the World Modeling Layer employs VLM to perform semantic compensation for missing or unreliable observations. This process mitigates world model mismatch caused by occlusion and constructs a scene representation that is more complete and stable than the raw outputs of YOLO, thereby improving the reliability of the world model used by the control system.

(3) Grounded Semantic–Geometric World Modeling Layer

The Grounded World Modeling Layer fuses incomplete geometric observations with the semantically compensated world model to form a state vector that more closely approximates the true physical scene. This unified representation serves as a common input for subsequent decision-making and motion planning modules, enabling the system to reason about target existence, occlusion states, and related conditions. As a result, this layer plays a central role in driving closed-loop grasping control under occluded scenarios.

(4) Decision and Task Planning Layer

The Decision and Task Planning Layer functions as a hybrid controller based on the cognitive state of the system. It treats the unified object set as the system’s belief state and switches between different operational modes—namely direct grasping and occlusion handling. Through this mode-switching mechanism, the system maintains closed-loop controllability of grasping tasks under occlusion and partial observability.

(5) Semantic-to-Motion Control Layer

The Semantic-to-Motion Control Layer acts as the continuous-space control executor of the system. It maps high-level cognitive states and discrete decision outcomes into physically executable end-effector poses and joint-level control commands. In doing so, semantic-level objectives are closed-loop grounded into real robotic motion through deterministic motion control.

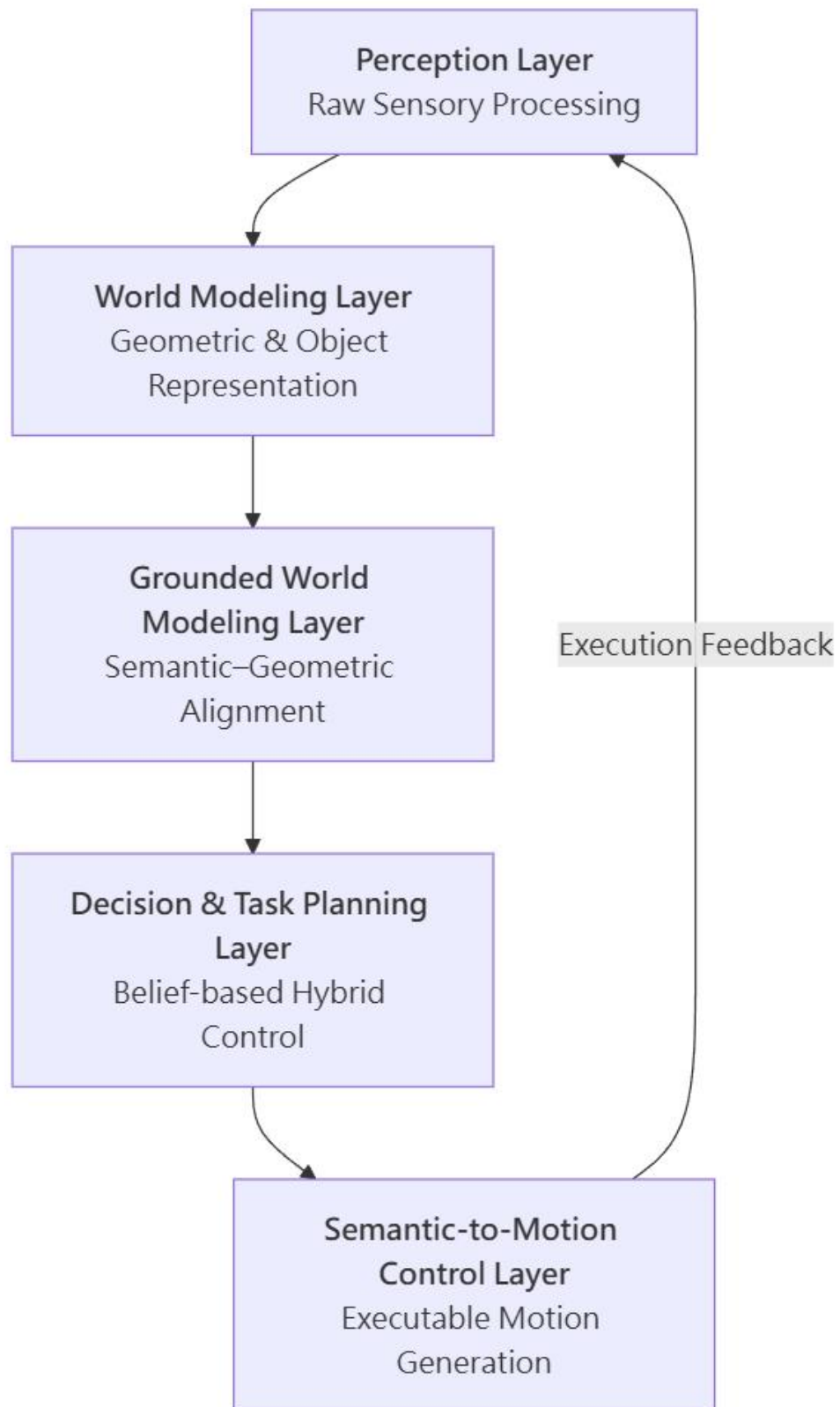


Fig. 3-2. Illustration of the Hierarchical Closed-Loop Control Architecture

3.1 Perception Layer

This section provides a detailed description of the visual perception module of the proposed system. Serving as the input interface of the overall architecture, the Perception Layer is responsible for transforming unstructured physical environments into structured representations that can be interpreted by the robotic system. The overall processing pipeline of this layer is illustrated in Fig. 3-3.

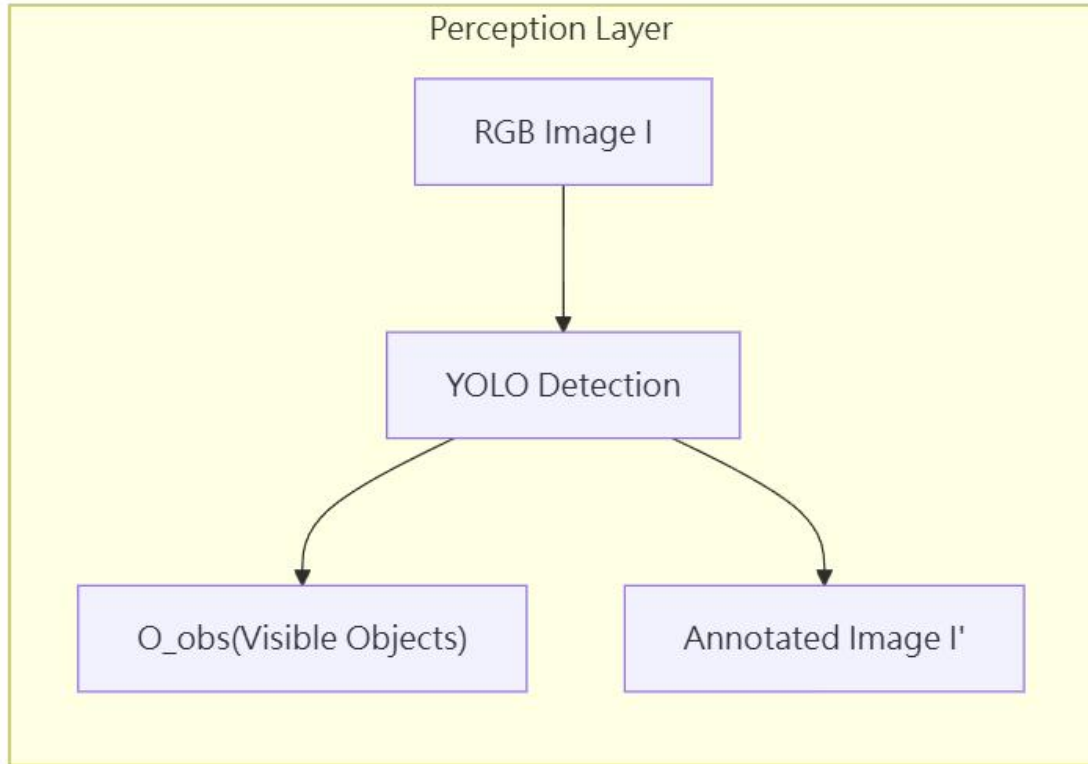


Fig. 3-3. Processing Pipeline of the Perception Layer

In the visual perception stage, the system acquires a three-channel RGB image of the current grasping scene with spatial resolution $H \times W$ through an RGB camera.

The system first applies the YOLO object detection network to the input image, producing a structured set of observed objects $\mathcal{O}_{\text{obs}}$:

$$\mathcal{O}_{\text{obs}} = \{ o_i | o_i = (b_i, p_i, c_i, l_i, s_i) \}_{i=1}^N \quad (3-1)$$

where

b_i : denotes the geometric bounding box parameters output by the detection model,

which are used for geometric reasoning and are visualized in the annotated image I' ;

p_i : denotes (x_i, y_i) the pixel coordinates of the bounding box center in the image coordinate system;

c_i : denotes the class label predicted by YOLO;

l_i : denotes the instance index, which is used to distinguish different object instances that share the same class label c_i , thereby enabling unique identification in multi-instance scenes;

s_i : denotes the detection confidence score, where $s_i \in [0, 1]$.

Meanwhile, YOLO generates an annotated image that is used as input for subsequent processing.

$$I' = \text{Annotate}(I, \mathcal{O}_{\text{obs}}) \quad (3-2)$$

The annotated image not only preserves the original visual information, but also serves as a form of visual prompt that provides attention guidance for the VLM-based Semantic Reasoning Layer. By explicitly highlighting object instances and their spatial features, the system guides the model to focus on local target regions and their spatial relationships, thereby improving its sensitivity to occlusion structures and physical configurations [29]. Through the explicit visualization of object spatial center points and instance indices in the image, the system enables the VLM to more accurately reason about occluded spatial relationships and object-level physical attributes.

3.2 World Modeling Layer

After obtaining the observed object set $\mathcal{O}_{\text{obs}}$ produced by the Perception Layer and the corresponding annotated image I' , the system enters the stage of semantic reasoning and world model construction. The objective of this stage is to compensate for instance segmentation failures that occur under occlusion, specular reflection, or low-illumination conditions, thereby correcting the system's world state

representation.

In occluded scenes, YOLO-based instance segmentation frequently suffers from missed detections. When a target object is severely occluded, its corresponding instance may be completely absent from $\mathcal{O}_{\text{obs}}$, causing the control system to incorrectly infer that the target does not exist and to prematurely terminate the grasping process.

This phenomenon arises from a model mismatch between the high-level world model and the true physical environment.

To address this issue, this study introduces a VLM as a semantic compensation module. As illustrated in Fig. 3-4, the overall architecture and data flow of the Semantic Reasoning Layer operate as follows: the structured detection results $\mathcal{O}_{\text{obs}}$, the annotated image I' , and a structured template prompt P (see Appendix A) are jointly provided as input to the VLM. The VLM infers objects that may exist in the scene but are not explicitly segmented by YOLO, along with their semantic attributes, thereby constructing a compensated semantic world model $\mathcal{O}_{\text{hyp}}$.

From a control-theoretic perspective, this process corresponds to state estimation under partial observability. Specifically, YOLO provides incomplete and noisy observations $\mathcal{O}_{\text{obs}}$, while the VLM functions as a high-level inference model that integrates visual context and task semantics to compensate for and regularize these observations. As a result, we obtain a more complete semantic world model $\mathcal{O}_{\text{hyp}}$ that offers a superior approximation of the true scene state.

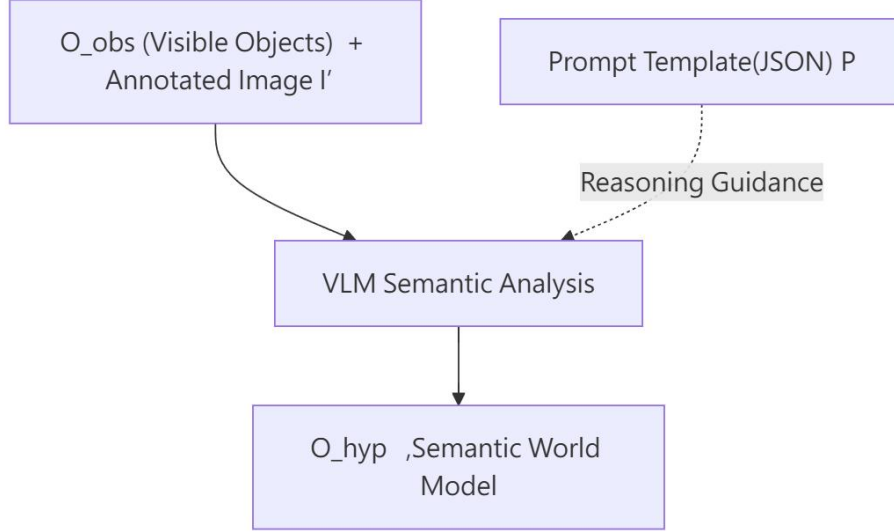


Fig. 3-4. VLM-Based Semantic World Model Inference under Partial Observability

At this stage, the system input and the resulting semantic inference process after multimodal information fusion can be formally expressed as:

$$(\mathcal{O}_{\text{obs}}, I', P) \xrightarrow{\text{VLM}} \mathcal{O}_{\text{hyp}} \quad (3-3)$$

where $\mathcal{O}_{\text{hyp}}$ denotes the semantic world model of the current scene inferred by the system, which is formally defined as:

$$\mathcal{O}_{\text{hyp}} = \{\hat{o}_j | \hat{o}_j = (\hat{c}_j, \hat{a}_j, \hat{s}_j^{\text{sem}})\}_{j=1}^M \quad (3-4)$$

$\hat{c}_j$: denotes the object class inferred through semantic reasoning, which may be corrected or completed beyond the raw visual detection results;

$\hat{a}_j$: denotes the semantic attributes and functional properties of the object (e.g., a cup is used for holding transparent liquids and primarily serves for drinking), which are used to indicate the object's potential role in the current manipulation task;

$\hat{s}_j^{\text{sem}}$: denotes the semantic confidence score produced by the internal reasoning mechanism of the VLM, which measures the reliability of the inferred semantic hypothesis for the object [30,31]. The semantic confidence $\hat{s}_j^{\text{sem}}$ is not a visual

classification probability, but a semantic matching score produced by the vision–language model under the current task context. The model jointly considers the object appearance, the scene context, and the manipulation objective to evaluate whether the inferred semantic class $\hat{c}_j$ and attributes $\hat{a}_j$ are consistent with the task requirements. The model assigns scores to multiple candidate semantic interpretations and normalizes them to obtain $\hat{s}_j^{\text{sem}}$, which reflects the relative reliability of the object as an effective manipulation target in the current task.

3.3 Grounded World Modeling Layer

When a robot arm operates under occluded conditions, relying solely on instance segmentation models in the Perception Layer often result in missed detections of target objects. Although the VLM in the World Modeling Layer can reasonably infer the existence and semantic attributes of occluded objects based on image context, LLMs have limited capability in handling precise spatial information such as bounding box coordinates and pixel-level positions.

To address this limitation, the proposed system introduces a Grounded World Modeling Layer between the semantic reasoning stage and the decision and control layers, as illustrated in Fig. 3-5. In this layer, the semantic world model $\mathcal{O}_{\text{hyp}}$ produced by the VLM is fused with the geometric observation set $\mathcal{O}_{\text{obs}}$ generated by YOLO into a unified and structured object representation. This fusion yields a belief state that more closely approximates the true physical scene and provides a stable input to the subsequent Decision and Task Planning Layer.

By grounding semantic inference in geometric observations, this layer reduces the reasoning burden on the language model and effectively improves both the stability and the responsiveness of the system’s decision-making process.

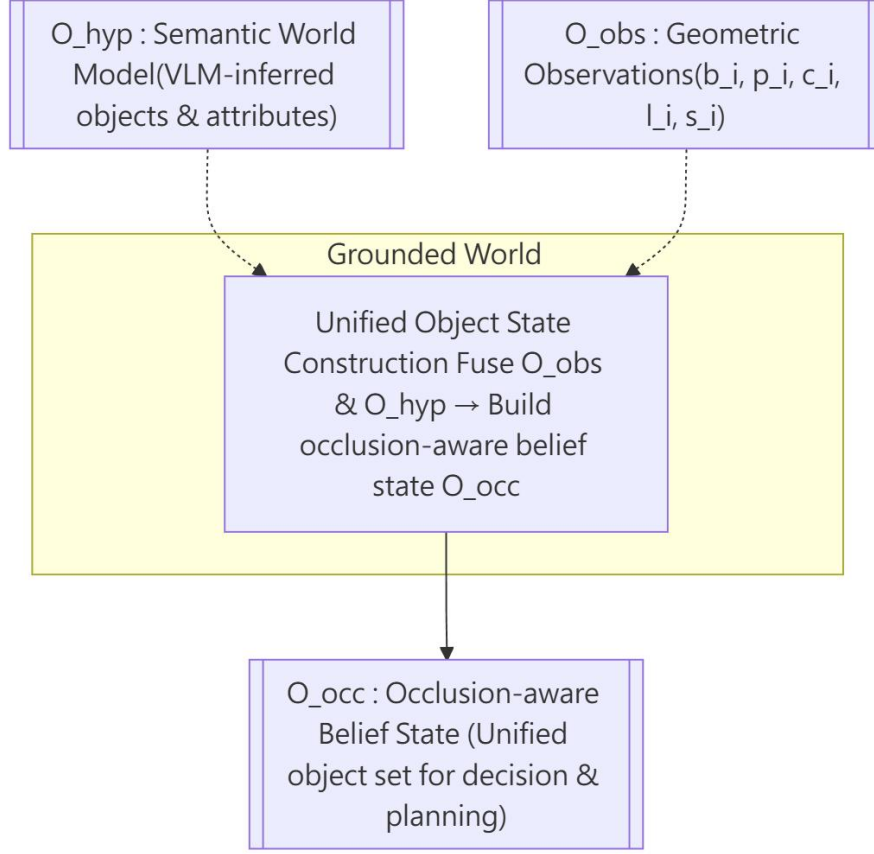


Fig. 3-5. Construction of the Belief State under Occluded Conditions

This layer takes as input the geometric observation set $\mathcal{O}_{\text{obs}}$ produced by the Perception Layer and the semantic world model $\mathcal{O}_{\text{hyp}}$ constructed by the Semantic Reasoning Layer. Rather than representing a deterministic mathematical function, the semantic–geometric alignment process serves as a fusion operator that integrates these two heterogeneous sources to construct a unified occlusion-aware object set $\mathcal{O}_{\text{occ}}$.

This operator performs instance-level association between semantic hypotheses and geometric detections, evaluates their spatial consistency, and reasons about potential occlusion relationships. Objects that are not directly observed but are semantically inferred are retained with associated confidence scores, while inconsistent or unreliable hypotheses are suppressed.

$$\mathcal{O}_{\text{occ}} = \mathcal{O}_{\text{obs}} \oplus \mathcal{O}_{\text{hyp}}: = \{\tilde{o}_k\}_{k=1}^K \quad (3-6)$$

Here, each $\tilde{o}_k$ represents a potentially existing object instance in the current belief state of the system. The fusion mechanism enables the system to simultaneously leverage geometric observation information and semantic inference results, forming a unified and complete input representation. This representation provides a stable data foundation for subsequent occlusion reasoning and high-level decision making, as formally described below:

$$\tilde{o}_k = \begin{cases} (b_k, p_k, \hat{c}_k, l_k, \hat{a}_k, \hat{s}_k^{\text{sem}}), & \text{if match} \\ (\phi_1, \phi_2, \hat{c}_k, l_k, \hat{a}_k, \hat{s}_k^{\text{sem}}), & \text{if unmatched} \end{cases} \quad (3-7)$$

Geometric Observation Parameters:

The tuple (b_k, p_k, l_k) represents the spatial data extracted from the perception layer, where:

b_k : denotes the bounding box of the object;

p_k : denotes the pixel coordinates corresponding to the center of the bounding box;

l_k : denotes the instance index, utilized to differentiate between distinct object instances.

ϕ_1 : denotes the null bounding box placeholder, indicating that no valid geometric region can be assigned to the semantic hypothesis;

ϕ_2 : the null pixel location placeholder, indicating that no valid image coordinate is available for the semantic hypothesis.

Semantic Reasoning Parameters:

The tuple $(\hat{c}_k, \hat{a}_k, \hat{s}_k^{\text{sem}})$ signifies the attributes inferred by the Semantic Reasoning Layer, where:

$\hat{c}_k$: denotes the inferred semantic class of the object;

$\hat{a}_k$: denotes the semantic attributes and functional properties associated with the object;

$\hat{s}_k^{\text{sem}}$: denotes the semantic confidence score indicating the reliability of the semantic inference.

Status:

Matched: Indicates that the semantic hypothesis object has been successfully associated with a geometric observation in the current perceptual set.

Unmatched: Indicates that the semantic hypothesis cannot be correlated with any available geometric observation. In this state, the geometric parameters are assigned null values.

3.4 Decision and Task Planning Layer

After semantic reasoning and grounded world modeling, the system obtains a unified object set $\mathcal{O}_{occ}$. This set integrates both the geometrically observable information provided by the Perception Layer and the potential target hypotheses inferred through semantic reasoning. As a result, $\mathcal{O}_{occ}$ can be regarded as the system's belief state representing the current scene.

Based on this belief state, the Decision and Task Planning Layer is responsible for formulating and solving the key control problems involved in robotic grasping under incomplete observations. Specifically, this layer performs decision reasoning with respect to the existence of the target object, its occlusion status, and the selection of feasible manipulation strategies. That is, it determines whether the target truly exists in the current scene, whether it is occluded by other objects, and which grasping or occlusion-handling strategy should be adopted under the current belief state.

From a control-theoretic perspective, these problems together constitute a high-level decision-making problem. Therefore, this study introduces the Decision and Task Planning Layer between the World Modeling Layer and the low-level motion control layer. Taking the unified state representation as input, this layer performs discrete judgments on target existence and occlusion status under the joint influence of task-level semantic constraints and geometric feasibility constraints, and accordingly selects an appropriate operation mode, such as directly grasping the target or prioritizing the removal of occluding objects.

As illustrated in Fig. 3-6, the Decision and Task Planning Layer first parses the

natural language task instruction into a structured semantic target representation, and verifies the existence of the target within the unified belief state. Once a valid target is detected, the system further performs occlusion reasoning to determine whether the target can be directly grasped or whether occluding objects must be removed beforehand. Finally, the system switches to the corresponding operation mode based on the decision results, and transforms the high-level decisions into executable control commands, which are then passed to the Semantic-to-Motion Control Layer.

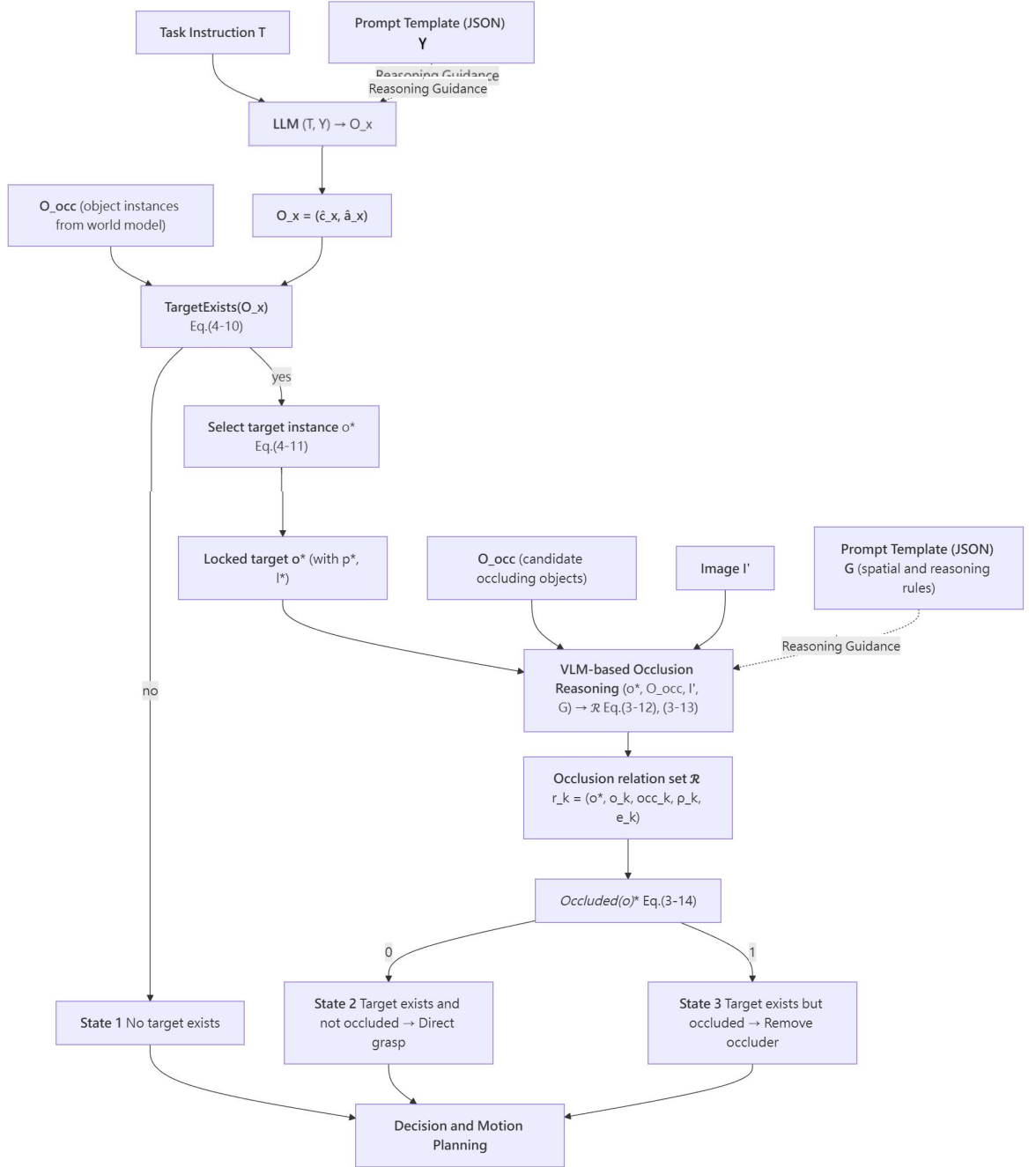


Fig. 3-6. Belief-State-Driven Occlusion-Aware Decision Pipeline

3.4.1 Target Selection

3.4.1.1 Target Definition and Task Semantics

Under the constraints of the prompt Y (see Appendix B), the LLM first performs target existence determination over the unified semantic candidate set $\mathcal{O}_{occ}$, in order to determine whether the current scene contains any graspable target that corresponds to the task instruction T . Based on this judgment, the system decides whether to enter the subsequent grasp planning stage.

•Under the constraints imposed by the prompt Y the high-level decision-making module utilizes the LLM to perform semantic parsing of the natural language task instruction T , and maps it into a structured representation of the task-level semantic target $\mathcal{O}_x$, as illustrated in Eq. (3-8):

$$(T, Y) \xrightarrow{\text{LLM}} \mathcal{O}_x \quad (3-8)$$

The task-level semantic objective is defined as:

$$\mathcal{O}_x = (\hat{c}_x, \hat{a}_x) \quad (3-9)$$

where

$\hat{c}_x$: denotes the target semantic category;

$\hat{a}_x$: represents the semantic attribute constraints of the target (e.g., attributes such as 'transparent liquid in a container' or 'brown liquid in a container').

3.4.1.2 Decision Function Logic

Subsequently, the system evaluates the existence of the target object based on the unified object set $\mathcal{O}_{occ}$, the target existence decision function is formulated as:

$$\text{TargetExists}(\mathcal{O}_x) = \begin{cases} 1, \exists \tilde{o}_m \in \mathcal{O}_{\text{occ}} \text{ s.t. } \hat{c}_m = \hat{c}_x \wedge \hat{a}_m \models \hat{a}_x \wedge \hat{s}_m^{\text{sem}} \geq \tau, \\ 0, \text{ otherwise} \end{cases} \quad (3-10)$$

where

$\hat{a}_m \models \hat{a}_x$: indicates that the semantic attributes of object instance $\tilde{o}_m$ satisfy the attribute constraints required by the task objective. For example, when the task requires an object to be capable of containing liquid, if an instance is inferred as a transparent cup that can hold water, it is regarded as satisfying this semantic constraint;

$\hat{s}_m^{\text{sem}}$: denotes the semantic confidence score, which measures the reliability of the semantic matching result (see Eq. (3-4));

τ : denotes the minimum confidence threshold, used to filter out low-confidence semantic hypotheses, thereby preventing erroneous decisions caused by unreliable semantic inference.

This design enables the system to eliminate semantic ambiguity in multi-instance scenes and to avoid erroneous decisions caused by objects that belong to the same category but do not satisfy the task-level semantic requirements. For example, when the task is “I want to drink water,” even if cup-class objects are present in the scene, the system will correctly determine that the target does not exist if all detected cups are semantically associated with containing coffee, and will therefore refrain from executing a grasp action.

3.4.1.3 Target Instance Selection

When the target existence decision yields a value of 1, the system proceeds to select a unique grasping target from the candidate set that satisfies the constraints defined in Eq. (3-10). If multiple candidate objects meet these conditions, the system applies a semantic-confidence-priority principle to select the final grasping target

from the set. The selection criterion is defined as follows:

$$o^* = \arg \max_{\tilde{o}_m} \{ \hat{s}_m^{sem} | \tilde{o}_m \text{ satisfies Eq. (3-10)} \} \quad (3-11)$$

where

o^* : denotes the selected target instance, which inherits the physical index l_m and spatial position p_m of the corresponding object, and serves as the unique referent in the subsequent occlusion verification and grasp decision process.

3.4.2 Target Occlusion Determination

After the existence of the target object has been confirmed (see Section 3.4.1), the system further proceeds to the target occlusion determination stage. In this stage, under the guidance of the prompt G , the system leverages a VLM to perform a joint semantic - geometric assessment of the occlusion relationship between the target instance and candidate occluding objects, and generates the corresponding grasping decision strategy based on the assessment results.

3.4.2.1 VLM-based Occlusion Reasoning for Target Verification

$$(o^*, \mathcal{O}_{occ}, I', G) \xrightarrow{\text{VLM}} R \quad (3-12)$$

where

o^* : denotes the currently selected grasp target instance see Eq. (3-11);

$\mathcal{O}_{occ}$: denotes the unified object set (see Section 3.3);

I' : represents the visually annotated image of the current scene (see Section 3.1);

G : denotes the prompt template that encodes spatial constraints and reasoning rules (see Appendix C);

R : denotes the set of occlusion reasoning results produced by the VLM given the input $(o^*, \mathcal{O}_{occ}, I', G)$, This result not only provides a judgment on whether the

target is occluded, but also supplies the corresponding semantic explanations, which are used to support subsequent decision-making and grasp strategy selection.

3.4.2.2 Structured Occlusion Reasoning Result Set $\mathcal{R}$

The occlusion reasoning result set $\mathcal{R}$ consists of a collection of candidate occlusion-relation tuples. Each element in the set is defined in the following form:

$$r_k = (o^*, o_k, occ_k, \rho_k, e_k). \quad (3-13)$$

where

o^* : denotes the currently selected grasp target instance see Eq. (3-11);

$o_k \in \mathcal{O}_{occ}$: denotes the k -th candidate occluding object;

$occ_k \in \{0,1\}$: denotes the binary occlusion judgment for the k -th candidate occluding object (1 for occluded, 0 otherwise);

$\rho_k \in [0,1]$: represents the semantic confidence of this occlusion judgment;

e_k : denotes the corresponding natural language explanation, such as the occlusion cause or spatial relationship description.

For example, when the target object is $o^* = cup$, the VLM may produce the following occlusion reasoning results: $r_1 = (o^* = cup, o_1 = spoon, \rho_1 = 0.92, e_1 = \textit{The spoon is positioned above the rim of the cup})$

The above occlusion reasoning results not only provide a binary judgment on whether the target object is occluded, but also output the corresponding semantic confidence and spatial explanation of the occlusion relationship. This enables the subsequent grasp strategy selection to be grounded in interpretable spatial evidence, rather than relying solely on an untraceable yes/no decision.

3.4.2.3 Definition of the Occlusion Decision Function

Let o^* denote the currently selected grasp target. The occlusion decision function is defined as follows:

$$\text{Occluded}(o^*) = \begin{cases} 1, \exists r_k \in \mathcal{R} \text{ s.t. } (\text{occ}_k = 1 \wedge \rho_k \geq \tau_{\text{vlm}}) \\ 0, \text{ otherwise} \end{cases} \quad (3-14)$$

This definition states that as long as at least one candidate occluding object o_k is judged to form a valid occlusion of the target object o^* (i.e., $\text{occ}_k=1$) and its confidence exceeds the threshold τ_{vlm} , the system determines that the target is occluded.

3.4.3 High-Level Decision-Making and Target Anchoring

3.4.3.1 Decision States and Discrete Control Modes

The system abstracts the current high-level cognitive state into a discrete state vector:

$$x_d = (\text{TargetExists}(\mathcal{O}_x), \text{Occluded}(o^*)) \in \{0,1\}^2 \quad (3-15)$$

where:

$\text{TargetExists}(\mathcal{O}_x)$: indicates whether a grasp target satisfying the semantic constraints exists in the current scene;

$\text{Occluded}(o^*)$: indicates whether the currently selected target instance o^* is in an occluded state.

This binary state vector maps complex continuous perceptual information—such as occlusion and semantic uncertainty into a low-dimensional, decidable discrete control state, thereby providing a stable and compact input representation for high-level decision making.

On this basis, the high-level operational strategy S of the system is modeled as a mode-switching controller, which switches between different operation modes according to the discrete state x_d :

$$S = \begin{cases} \text{Abort,} & \text{if TargetExists}(\mathcal{O}_x) = 0 \\ \text{Grasp Target,} & \text{if TargetExists}(\mathcal{O}_x) = 1 \wedge \text{Occluded}(o^*) = 0 \\ \text{Remove Occluded,} & \text{if TargetExists}(\mathcal{O}_x) = 1 \wedge \text{Occluded}(o^*) = 1 \end{cases} \quad (3-16)$$

The above control strategy S maps the discrete cognitive state obtained from perception and semantic reasoning into concrete operation modes, thereby forming a high-level mode-switching controller driven by target existence and occlusion status.

When the system determines that no grasp target satisfying the task semantics exists in the current scene, the controller switches to the Abort mode to avoid meaningless grasp attempts.

When the target exists and is not occluded, the system directly enters the GraspTarget mode to execute the grasp.

When the target exists but is occluded, the controller switches to the Remove Occluder mode, prioritizing the removal of occluding objects in order to restore the graspability of the target.

Through this discrete state - based mode-switching mechanism, the system can explicitly distinguish between directly graspable scenarios and those requiring occlusion handling, thereby improving the stability and reliability of grasp decision-making in cluttered environments.

3.4.3.2 Execution Object Anchoring

After the control mode is determined, the system further selects a concrete execution object $\hat{o}$ from the unified object set $\mathcal{O}_{occ}$, so as to establish the anchoring from high-level semantic decisions to physical execution.

When the system is in the direct grasp mode, the execution object $\hat{o}$ is set to the target object o^* .

When the system is in the Remove Occluder mode, $\hat{o}$ is selected from the set of objects that have been identified as valid occluders, and the one with the highest

semantic confidence is chosen as the execution object.

$$\hat{o} = \begin{cases} o^*, & \text{if } S = \text{Grasp Target} \\ \arg \max_{o_k} \{ \rho_k | \text{occ}_k = 1 \}, & \text{if } S = \text{Remove Occluder} \end{cases} \quad (3-17)$$

where

ρ_k : denotes the confidence that object o_k acts as a valid occluder;

occ_k : denote the corresponding occlusion indicator.

Once the execution object $\hat{o}$ is determined, the system can directly retrieve its geometric attributes stored in the object tuple, and obtain the pixel coordinate p^* of the target as the input to the low-level motion control module.

3.4.3.3 Closed-Loop Grasping Control under Occlusion

After the system executes the Remove Occluder action, the geometric structure of the scene changes. Consequently, the visual perception and semantic reasoning modules are re-triggered, and the current scene state variables $\text{TargetExists}(\mathcal{O}_x)$ and $\text{Occluded}(o^*)$ are updated accordingly.

Under the discrete control logic defined in Eq. (3-16), this process forms a closed-loop control mechanism: the system continuously iterates through the cycle of perception, reasoning, decision and execution until the final grasping condition is satisfied:

$$\text{TargetExists}(\mathcal{O}_x) = 1 \wedge \text{Occluded}(o^*) = 0$$

That is, once the target exists and is no longer occluded, the system transitions into the final target grasping phase.

3.5 Semantic-to-Motion Control Layer

The semantic-to-motion control layer is responsible for transforming the outputs of high-level semantic reasoning and discrete decision making into continuous motion control commands that can be directly executed by a robot arm. It serves as a critical closed-loop bridge connecting language understanding, world modeling, and physical execution.

From a control-theoretic perspective, this layer performs discrete-to-continuous grounding, i.e., it maps high-level cognitive decisions in a discrete space into feasible control actions in the continuous domain. Through this process, abstract cognitive states and strategy selections are instantiated as joint-level motion trajectories that satisfy physical and kinematic constraints. The core function of this layer is to progressively translate symbolic semantic decisions into control variables with explicit geometric and dynamic meanings, while enforcing constraint checks to ensure feasibility and safety in real-world execution.

Therefore, this layer constitutes the control mapping from cognition to physical actuation within the overall system, and forms a central component for achieving closed-loop intelligent control.

Specifically, the system first projects the target pixel coordinates inferred by the semantic reasoning module into a spatial target point in the robot coordinate frame, and then computes the corresponding joint angles via inverse kinematics. The resulting motion plan is subsequently validated in a simulation environment. If either the inverse kinematics solution or the simulation-based feasibility check fails, the grasp is judged to be physically non-executable, and the system aborts the action to avoid unreachable or unsafe motions. Once the plan is verified, the joint commands are further converted into PWM control signals that can be directly executed by the embedded controller.

3.5.1 Binding Semantic Objects to Geometric Targets

This subsection aims to establish a mapping mechanism from the high-level discrete semantic decision space to the physical execution space. By binding the high-level control strategy S to a concrete physical object instance, the system can provide a clear execution target for the robot arm. Once the high-level control strategy S is determined, the system selects a specific execution object $\hat{o}$ from the unified object set according to Eq. (3–17).

Case 1: Direct Grasp Mode ($S=\text{Grasp Target}$)

In this mode, the execution object $\hat{o}$ is directly mapped to the target instance o^* .

Here, o^* is the unique target selected by the target selection function defined in Eq. (3–11) from the candidate set that satisfies the geometric constraints, based on the highest semantic confidence. This mapping process ensures that the robot end-effector acts directly on the core object of the task.

Case 2: Occluder Removal Mode ($S=\text{Remove Occluder}$)

When physical occlusion exists in the current environment, the system switches to the occlusion-handling logic. In this mode, the system iterates through all candidate objects in the unified object set $\mathcal{O}_{\text{occ}}$. First, valid occluders are filtered by the occlusion indicator $\text{occ}_k = 1$. Then, the object with the highest occlusion confidence ρ_k is selected as the current execution object $\hat{o}$.

3.5.2 Geometric Coordinate Extraction and Spatial Projection

3.5.2.1 Pixel Coordinate Decoding and Distortion Correction

This subsection maps the image location of the selected object to a three-dimensional spatial target point that can be interpreted by the robot.

The system selects an execution object $\hat{o}$ from the unified object set $\mathcal{O}_{\text{occ}}$ constructed by fusing geometric observations $\mathcal{O}_{\text{obs}}$ and semantic hypotheses $\mathcal{O}_{\text{hyp}}$ (Eq. (3 – 6)). If $\hat{o}$ is matched to a geometric observation, its bounding box center $p_d = (u_d, v_d)$ is taken from the geometric parameters (b_k, p_k, l_k) inherited from the perception layer (Eq. (3 – 7)).

To eliminate the influence of lens distortion on localization accuracy, the intrinsic

camera matrix $\mathbf{K}$ and the distortion parameter set δ are used to correct the raw pixel coordinates, yielding the undistorted pixel position (u, v) :

$$(u, v) = \text{undistort}(u_d, v_d, \mathbf{K}, \delta) \quad (3-18)$$

where:

(u_d, v_d) : denotes the raw distorted pixel coordinates extracted from the original image;

$\mathbf{K}$: denotes the camera intrinsic matrix representing the internal geometric parameters of the camera;

δ : denotes the camera distortion parameter set;

(u, v) : denotes the ideal pixel coordinates after distortion correction that conform to the pinhole camera model.

The intrinsic camera matrix $\mathbf{K}$ is defined as:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (3-19)$$

where:

f_x, f_y : denote the focal lengths (in pixel units) along the horizontal and vertical image axes;

(c_x, c_y) : denotes the coordinates of the principal point, representing the intersection of the optical axis with the image plane.

3.5.2.2 Ray Projection from Image to Camera Coordinate System

To reconstruct the spatial relationship, the distortion-corrected pixel coordinates are projected into the camera coordinate frame C as a normalized 3D ray direction vector $\mathbf{d}_C$. This transformation is performed using the inverse of the camera intrinsic matrix $\mathbf{K}^{-1}$ as follows:

$$\mathbf{d}_C = \begin{bmatrix} x_n \\ y_n \\ 1 \end{bmatrix} = \mathbf{K}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (3-20)$$

The intrinsic matrix inverse $\mathbf{K}^{-1}$ is derived from the camera's internal parameters:

$$\mathbf{K}^{-1} = \begin{bmatrix} \frac{1}{f_x} & 0 & -\frac{c_x}{f_x} \\ 0 & \frac{1}{f_y} & -\frac{c_y}{f_y} \\ 0 & 0 & 1 \end{bmatrix} \quad (3-21)$$

By substituting Eq. (3-21) into Eq. (3-20), the normalized image coordinates x_n , y_n are explicitly expressed as:

$$x_n = \frac{u-c_x}{f_x}, \quad y_n = \frac{v-c_y}{f_y} \quad (3-22)$$

where:

$\mathbf{d}_C$: denotes the 3D ray direction vector in the camera coordinate system;

(u, v) : denotes the undistorted pixel coordinates obtained after removing lens distortion, representing the point's position in the 2D image plane (in pixels);

(x_n, y_n) : defined on the normalized image plane at $Z = 1$ in the camera coordinate system;

f_x, f_y : denotes the effective focal lengths of the camera lens along the x and y axes, respectively;

(c_x, c_y) : denotes the coordinates of the principal point, see Eq. (3-19).

3.5.2.3 Hand – Eye Calibration and Coordinate Transformation

Since the system adopts an eye-in-hand configuration[32], the camera is mounted on the robot end-effector. The homogeneous transformation matrix from the camera coordinate frame C to the robot base frame B is given by

$${}^B\mathbf{T}_C(\theta) = {}^B\mathbf{T}_E(\theta) \cdot {}^E\mathbf{T}_C(\theta) = \begin{bmatrix} \mathbf{R}_{BC}(\theta) & \mathbf{t}_{BC}(\theta) \\ 0 & 1 \end{bmatrix} \quad (3-23)$$

where:

${}^E\mathbf{T}_C$: denotes the fixed hand–eye calibration matrix from the camera frame to the end-effector frame, obtained through calibration;

${}^B\mathbf{T}_E(\theta)$: denotes the real-time forward kinematics transformation from the end-effector frame to the robot base frame, computed from the joint angles;

${}^B\mathbf{T}_C(\theta)$: denotes the composed homogeneous transformation from the camera frame to the robot base frame, given by

$${}^B\mathbf{T}_C(\theta) = \begin{bmatrix} \mathbf{R}_{BC}(\theta) & \mathbf{t}_{BC}(\theta) \\ \mathbf{0} & 1 \end{bmatrix},$$

where $\mathbf{R}_{BC}(\theta)$ and $\mathbf{t}_{BC}(\theta)$ represent the rotation matrix and translation vector of the camera frame relative to the base frame, respectively.

Furthermore,

(1)Forward Kinematics and Hand–Eye Calibration

The forward kinematics from the end-effector to the base frame is written as:

$${}^B\mathbf{T}_E(\theta) = \begin{bmatrix} \mathbf{R}_{BE}(\theta) & \mathbf{t}_{BE}(\theta) \\ \mathbf{0}^T & 1 \end{bmatrix} \quad (3-24)$$

where:

$\mathbf{R}_{BE}(\theta)$: denotes the rotation matrix from frame E to frame B, computed from the robot joint angles;

$\mathbf{t}_{BE}(\theta) = [t_x(\theta), t_y(\theta), t_z(\theta)]^T$: denotes the position of the end-effector origin in the base frame.

The fixed hand–eye transformation from the camera to the end-effector frame is given by

$${}^E\mathbf{T}_C = \begin{bmatrix} \mathbf{R}_{EC} & \mathbf{t}_{EC} \\ \mathbf{0}^T & 1 \end{bmatrix} \quad (3-25)$$

where:

$\mathbf{R}_{EC}$: denotes the constant rotation matrix describing the camera's orientation

relative to the end-effector;

$\mathbf{t}_{EC} = [t_x^{EC}, t_y^{EC}, t_z^{EC}]^T$ denotes the translation vector from the end-effector origin to the camera optical center.

(2) Translation vector from the end-effector origin to the camera optical center.

Substituting the above expressions into Eq. (3-23), the camera-to-base transformation can be expanded as:

$${}^B\mathbf{T}_C(\theta) = \begin{bmatrix} \mathbf{R}_{BE}(\theta)\mathbf{R}_{EC} & \mathbf{R}_{BE}(\theta)\mathbf{t}_{EC} + \mathbf{t}_{BE}(\theta) \\ \mathbf{0}^T & 1 \end{bmatrix} \quad (3-26)$$

Thus, the rotation and translation components are given by

$$\mathbf{R}_{BC}(\theta) = \mathbf{R}_{BE}(\theta)\mathbf{R}_{EC} \quad (3-27)$$

$$\mathbf{t}_{BC}(\theta) = \mathbf{R}_{BE}(\theta)\mathbf{t}_{EC} + \mathbf{t}_{BE}(\theta) \quad (3-28)$$

where:

$\mathbf{R}_{BE}(\theta)$: denotes the real-time rotation matrix of the end-effector with respect to the robot base, fully describing the orientation of the end-effector under joint configuration θ ;

$\mathbf{R}_{EC}$: denotes the fixed rotation offset between the camera coordinate frame and the robot tool (end-effector) frame, obtained by hand-eye calibration;

$\mathbf{t}_{EC}$: denotes the fixed translation vector describing the physical mounting displacement of the camera relative to the end-effector;

$\mathbf{R}_{BC}(\theta)$ and $\mathbf{t}_{BC}(\theta)$: jointly define the complete geometric mapping from the camera coordinate frame to the robot base coordinate frame.

(3) Ray Transformation to the Base Frame

Finally, the ray direction vector is rotated into the base coordinate frame B using the rotation matrix $\mathbf{R}_{BC}$:

$$\mathbf{d}_B = \mathbf{R}_{BC}(\theta) \mathbf{d}_C \quad (3-29)$$

where:

$\mathbf{d}_B$: ray direction vector in the base frame after back-projection and coordinate transformation.

$\mathbf{R}_{BC}(\theta)$: defined in Eq. (3-27);

$\mathbf{d}_C$: defined in Eq. (3-20).

This transformation establishes the geometric bridge between visual perception and robot motion planning, enabling consistent spatial reasoning and closed-loop control.

3.5.2.4 Depth Recovery from a Planar Constraint

In the absence of explicit depth information, a planar constraint is introduced by assuming that the target lies on a known workspace plane at a fixed height $z = Z_{\text{table}} = 330\text{mm}$, which allows the depth scale λ^* to be recovered.

The 3D ray in the robot base coordinate frame B is parameterized as:

$$\mathbf{p}_B(\lambda) = \mathbf{o}_B + \lambda \mathbf{d}_B \quad (3-30)$$

where:

λ : ray scale parameter (depth scale factor) along direction $\mathbf{d}_B$;

$\mathbf{p}_B$: 3D point on the ray in the robot base frame for a given λ ;

$\mathbf{d}_B$: ray direction vector defined in Eq. (3-29);

$\mathbf{o}_B$: 3D position of the camera optical center in the base frame, given by

$$\mathbf{o}_B = \mathbf{t}_{BC}(\theta)$$

which corresponds to the translation component of the homogeneous transformation ${}^B\mathbf{T}_C(\theta) = {}^B$ in Eq. (3-23).

By applying the plane-height constrain $z = Z_{\text{table}}$, the ray scale parameter λ is obtained as:

$$\lambda^* = \frac{Z_{\text{table}} - o_z}{d_z} \quad (3-31)$$

where:

λ : recovered ray scale parameter at the intersection with the workspace plane;

Z_{table} : fixed height of the workspace plane in the base coordinate frame;

o_z : z-component of the camera center $\mathbf{o}_B$;

d_z : z-component of the ray direction vector $\mathbf{d}_B$.

Under this planar assumption, the 3D position of the target in the robot base frame is then computed as:

$$\mathbf{P}^* = \mathbf{o}_B + \lambda^* \mathbf{d}_B = [x^*, y^*, z^*]^T \quad (3-32)$$

where

λ^* : defined in Eq. (3-31);

$\mathbf{o}_B$, $\mathbf{d}_B$: have the same physical meaning as in Eq. (3-30);

x^*, y^*, z^* denote the Cartesian coordinates of the recovered target position expressed in the robot base coordinate frame B .

3.5.3 Five-DOF Inverse Kinematics and Feasibility Check

After obtaining the physical target position $\mathbf{P}^*$, the system maps it into the joint-space configuration of the robotic arm: $\theta = [\theta_1, \dots, \theta_5]^T$.

In the physical structure shown in Fig. 3-7, the end-effector includes a fixed wrist-to-gripper offset l_4 . In this formulation, the physical target position $\mathbf{P}^*$ represents the wrist center after compensating this fixed offset along the tool approach direction. This ensures geometric consistency between the kinematic model and the actual hardware structure.

This section derives the inverse kinematics for the 5-DOF manipulator shown in Fig. 3-7. In the adopted kinematic structure, joints $\theta_1 - \theta_4$ determine the spatial pose of the end-effector required for grasping, whereas joint θ_5 only drives the gripper

oscillation (yaw motion about the Z_5 axis) and does not affect the Cartesian position of the grasp point.

Therefore, θ_5 is excluded from the position-level inverse kinematics derivation and is treated separately in the grasp execution stage.

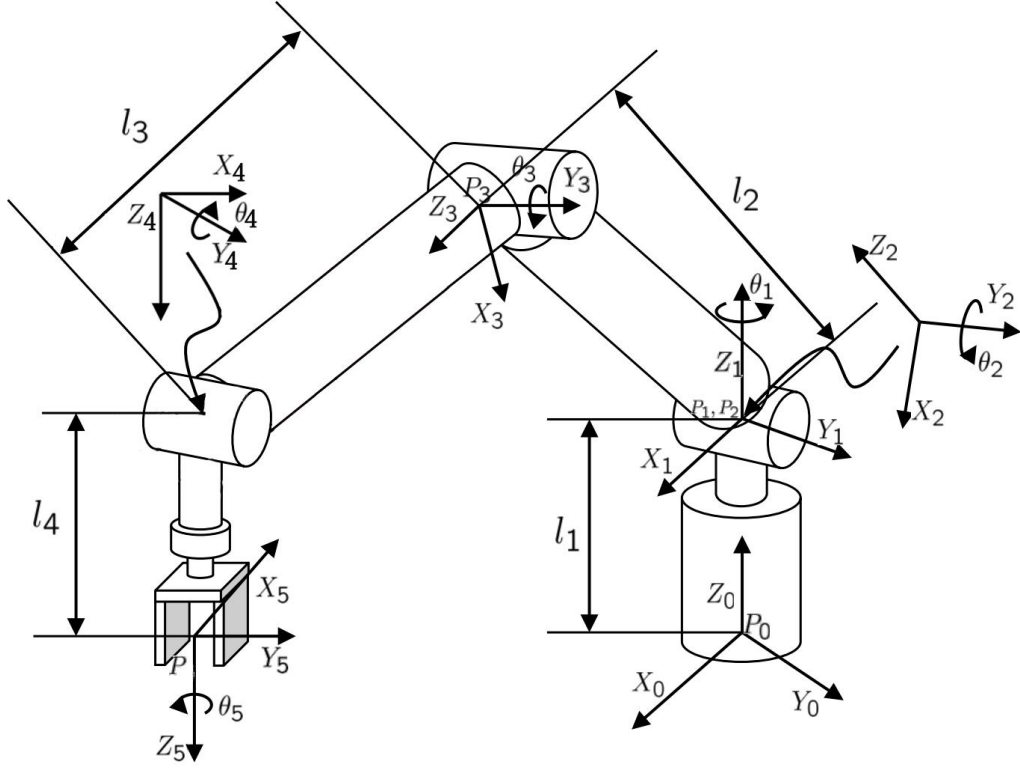


Fig. 3-7. Physical Implementation of the Proposed Robotic System

To facilitate the geometric derivation of the inverse kinematics, the spatial target position $P^* = [x^*, y^*, z^*]^T$ is first projected onto the planar kinematic chain formed by joints θ_2 and θ_3 .

The horizontal radial distance from the base axis and the vertical displacement relative to the shoulder joint are defined as:

$$r = \sqrt{(x^*)^2 + (y^*)^2} \quad (3-33)$$

$$z' = z^* - l_1 \quad (3-34)$$

Where l_1 is the vertical offset from the base frame origin to the shoulder joint axis (as labeled in Fig. 3-7). The remaining two links in the main planar chain are

denoted as l_2 (upper arm) and l_3 (forearm).

3.5.3.1 Base Joint Angle θ_1

The first joint θ_1 performs a rotation about the base axis Z_0 . Its purpose is to align the projection of the spatial target point onto the base plane X_0-Y_0 with the sagittal plane of the subsequent planar mechanism formed by joints θ_2 and θ_3 . Let the target position in the base frame be $\mathbf{P}^* = [x^*, y^*, z^*]^T$ as defined in Eq. (3-32). The projection of $\mathbf{P}^*$ onto the base plane X_0-Y_0 is given by (x^*, y^*) .

From planar geometry, the required base rotation angle is obtained as:

$$\theta_1 = \text{atan2}(y^*, x^*) \quad (3-35)$$

which uniquely determines the rotation about the vertical axis Z_0 .

After applying this rotation, the target point is aligned with the plane of motion of joints θ_2 and θ_3 , thereby reducing the spatial problem to a two-dimensional inverse kinematics subproblem in the (r, z') plane (see Eq. (3-34)).

3.5.3.2 Shoulder Joint θ_2

After applying θ_1 , the spatial problem reduces to a planar 2-link inverse kinematics problem in the (r, z') plane. The geometry is illustrated by a triangle formed by links l_2 , l_3 and the target point.

The auxiliary angle is first defined as:

$$\phi = \text{atan2}(z', r) \quad (3-36)$$

which represents the direction from the shoulder joint to the target point in planar coordinates. Once the elbow angle θ_3 is obtained (Sec. 3.5.3.3), the angle between link l_2 and the line connecting the shoulder to the target is given by

$$\psi = \text{atan2}(l_3 \sin \theta_3, l_2 + l_3 \cos \theta_3) \quad (3-37)$$

From planar triangle decomposition, the shoulder joint angle is therefore

$$\theta_2 = \phi - \psi \quad (3-38)$$

3.5.3.3 Elbow Joint θ_3

Consider the planar triangle formed in the (r, z') plane by the two link lengths l_2 and l_3 , and the distance from the shoulder joint to the target point. The Euclidean distance between the shoulder joint and the target projection is defined as:

$$s = \sqrt{r^2 + (z')^2} \quad (3-39)$$

Applying the law of cosines to this triangle yields

$$\cos \theta_3 = \frac{s^2 - l_2^2 - l_3^2}{2l_2l_3} = \frac{r^2 + (z')^2 - l_2^2 - l_3^2}{2l_2l_3} \quad (3-40)$$

For compact notation, define

$$D = \frac{r^2 + (z')^2 - l_2^2 - l_3^2}{2l_2l_3} \quad (3-41)$$

The elbow joint angle therefore satisfies

$$\theta_3 = \text{atan2}(\pm \sqrt{1 - D^2}, D) \quad (3-42)$$

In this work, the elbow-down configuration is adopted to ensure a collision-free and mechanically feasible posture.

A valid solution exists only if $|D| \leq 1$, which defines the reachable workspace of the planar linkage. If $|D| > 1$, the target position lies outside the reachable domain.

In practical implementation, the appropriate branch is selected according to joint limits and feasibility constraints, such as collision avoidance or proximity to the

current joint configuration to ensure smooth motion continuity.

3.5.3.4 Wrist Pitch θ_4

In the planar kinematic chain (r, z') plane, the orientation of the forearm relative to the shoulder joint is determined by the accumulated rotation of joints θ_2 and θ_3 . Therefore, the pitch angle of the forearm with respect to the base plane can be expressed as $\beta = \theta_2 + \theta_3$.

Let α denote the desired end-effector pitch angle in the planar chain, defined with respect to the base horizontal plane. The wrist joint θ_4 is compensating for the accumulated forearm rotation such that the overall tool pitch equals α . Hence, the orientation constraint is given by $\theta_2 + \theta_3 + \theta_4 = \alpha$.

which yields

$$\theta_4 = \alpha - (\theta_2 + \theta_3) \quad (3-43)$$

In the common case where a vertical tool approach is adopted (i.e., $\alpha = 0$ under the chosen orientation convention), the expression simplifies to

$$\theta_4 = -(\theta_2 + \theta_3) \quad (3-44)$$

3.5.3.5 Gripper Swing θ_5

Joint θ_5 rotates about axis Z_5 and provides the necessary degree of freedom for rotational alignment. While it is excluded from the position-level IK derivation due to its lack of influence on the grasp point coordinates, its value is dynamically assigned during the grasp execution stage to satisfy orientation constraints.

3.5.3.6 Closed-Form IK Summary and Feasibility Verification

The inverse kinematics of the proposed 5-DOF manipulator admits a closed-form analytical solution derived in Sec. 3.5.3.1–3.5.3.5.

Given the target position $\mathbf{P}^*$, the joint configuration is computed sequentially as

$$\theta^* = [\theta_1^*, \theta_2^*, \theta_3^*, \theta_4^*, \theta_5^*]^T \quad (3-45)$$

where

θ_1^* : planar projection on the base frame (Eq. 3-35),

θ_3^* : cosine law formulation (Eq. 3-42),

θ_2^* : planar triangle decomposition (Eq. 3-38),

θ_4^* : compensates the accumulated pitch (Eq. 3-43),

θ_5^* : grasp orientation.

To ensure the physical realizability of the computed configuration, the analytical solution is subjected to multi-level feasibility verification before command execution.

A solution is considered valid only if

$$\text{Feasible} = \begin{cases} 1, & \text{if } (\theta_i^{\min} \leq \theta_i^* \leq \theta_i^{\max}) \wedge (|D| \leq 1) \\ 0, & \text{otherwise} \end{cases} \quad (3-46)$$

where:

$\theta_i^{\min}, \theta_i^{\max}$: joint limit constraints for the i -th joint;

$|D| \leq 1$: guarantees geometric reachability, the condition collectively ensures that the target lies within the reachable workspace.

3.5.4 Embedded Execution Mapping

The system first computes the corresponding five-joint configuration θ^* from the target spatial position $\mathbf{P}^*$ using Eq. (3-32), and then verifies its physical feasibility via Eq. (3-46).

When Feasible=1, cases such as joint limit violations or unreachable targets are excluded. Only after successful validation is the joint command transmitted to the

low-level controller for execution.

The validated joint angles are mapped to PWM control signals for the embedded actuator drivers as:

$$\text{PWM}_{\text{target}}(\theta_i^*) = \text{PWM}_{\text{min},i} + \frac{\theta_i^* - \theta_i^{\text{min}}}{\theta_i^{\text{max}} - \theta_i^{\text{min}}} (\text{PWM}_{\text{max},i} - \text{PWM}_{\text{min},i}) \quad (3-47)$$

where:

$\theta_i^{\text{min}}, \theta_i^{\text{max}}$: physical joint angle limits;

$\text{PWM}_{\text{max},i}, \text{PWM}_{\text{min},i}$: minimum and maximum driving pulse widths for the i -th joint actuator.

3.6 Summary of the Proposed Control Framework

This chapter addresses the problem of robot arm grasping under occluded environments by constructing a hierarchical closed-loop control framework that spans from perception to execution. The system first transforms raw visual inputs into a structured object set through the perception and Semantic Reasoning Layers, and further derives an operable scene representation via a grounded semantic–geometric world modeling mechanism. On this basis, the complex continuous perceptual state is abstracted into a discrete control state.

Using this discrete state representation, the system can switch among three operational modes—abort grasping, direct target grasping, and occluder removal—with the control strategy defined in Eq. (3–16) providing explicit responses to the current scene state.

Furthermore, the Semantic-to-Motion Control Layer maps the discrete semantic decisions into the continuous physical control space. The system retrieves the target pixel coordinates from the geometric attributes stored in the object tuples and projects them into the robot base coordinate frame using camera calibration and hand–eye calibration models. The spatial target point is then converted into a joint configuration

that satisfies kinematic constraints through inverse kinematics solving and simulation-based feasibility verification. If no executable solution is found, the grasp is classified as physically infeasible, thereby preventing unsafe or failed actions.

Through the above design, this chapter establishes a complete perception–semantics–decision–motion–execution closed-loop control pipeline. This enables the robot arm to progressively converge toward a graspable target under incomplete observations by actively removing occluders when necessary. The proposed framework not only ensures consistency between semantic reasoning and physical execution, but also provides a structured control foundation for the experimental validation of system stability and robustness in complex occluded environments.

4 Experimental and Results Analysis

This chapter systematically validates the proposed occlusion-aware semantic decision and control framework in complex scenes through both qualitative and quantitative experiments.

The evaluation focuses on the correctness of target selection and the reliability of closed-loop decision-making, rather than the numerical precision of low-level actuator execution. We emphasize the stability and consistency of the five-layer closed-loop control architecture—perception, semantic reasoning, world modeling, decision, and motion control—under partial observability. To avoid the interference of hardware noise with high-level decision evaluation, the experiments primarily assess whether the system can select appropriate strategies and exhibit correct closed-loop behaviors under different occlusion and visual conditions. Particular attention is paid to the correctness of high-level decision-making when the robot operates under perceptual incompleteness and occlusion uncertainty. The experimental evaluation consists of three main components:

- (1) Experimental platform and hardware configuration;
- (2) Qualitative scenario-based experimental design and analysis;
- (3) Quantitative experimental design and analysis under varying condition parameters.

4.1 Experimental Platform and System Configuration

The proposed experimental platform is jointly composed of a host computing system and a robot arm execution unit, forming a closed-loop control pipeline that spans from high-level semantic decision-making to low-level physical execution. The physical deployment and hardware interconnection of the system are illustrated in Fig. 3-1. This architecture effectively supports robot arm grasping tasks in complex real-world environments.

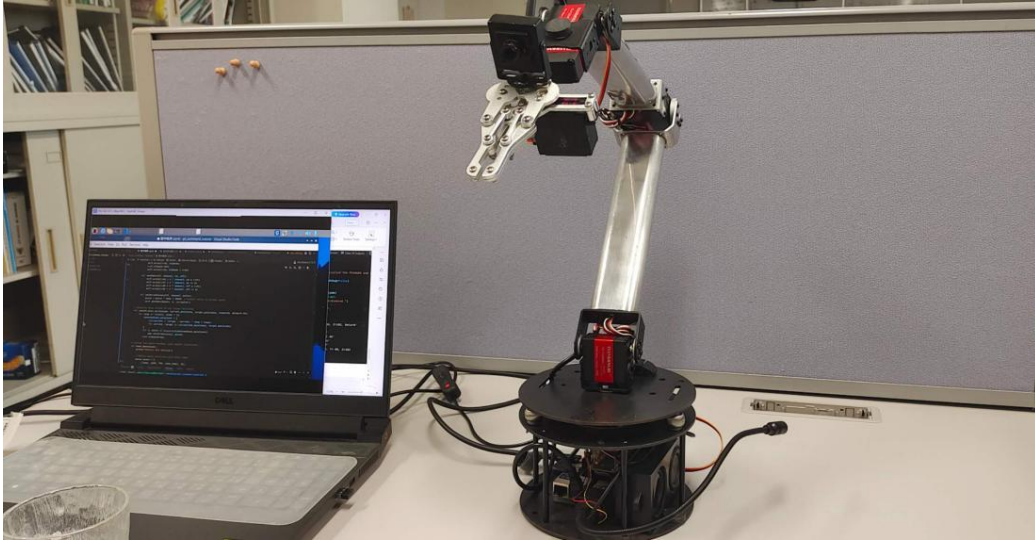


Fig. 4-1. Physical Implementation of the Proposed Robotic System

4.1.1 Actuator Design

The robot arm adopts a standard serial configuration, with each joint driven by a high-torque RDS5160 servo motor, ensuring high precision and reliability in the movement of each joint. The system is able to precisely control the motion of each servo motor through PWM signals sent by the lower-level controller (Raspberry Pi). The design of the robot arm incorporates a modular structure, with each joint's servo motor installed within a sturdy and lightweight frame, capable of providing stable and efficient power output under high loads.

4.1.2 Experimental Platform Computer System

The proposed experimental system adopts a hierarchical host-controller architecture, as illustrated in Fig. 4-2. This architecture establishes a closed-loop control pipeline from high-level semantic decision-making to low-level physical execution, and is well suited for practical robot arm grasping tasks.

The host computer runs on a workstation equipped with an Intel i9 series processor and an NVIDIA RTX series GPU. It is responsible for executing the Perception Layer, Semantic Reasoning Layer, World Modeling Layer, Grounded World Modeling Layer and Decision and Task Planning Layer defined in Chapter 3.

On the host side, the system performs object detection and semantic parsing on the input images, constructs a unified occlusion-aware world model, and generates high-level operational modes according to the current discrete state, such as direct target grasping or occluder removal.

The lower-level controller is implemented on a Raspberry Pi 5 and serves as the execution unit of the semantic-to-motion control layer. It receives the selected target object and its corresponding spatial goal from the host computer, computes the joint angles of the robot arm through inverse kinematics, and converts them into PWM control signals, which then drive the 5-DOF robot arm to perform grasping or occluder removal actions.

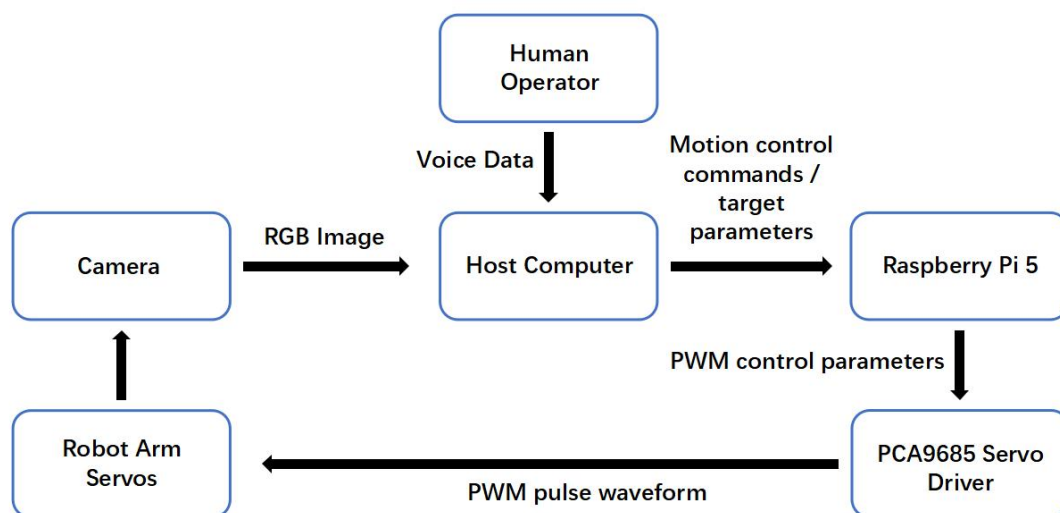


Fig. 4-2. Actual Operating Architecture of the Robot Arm Control System

4.2 Experiment I: Scenarios under Occluded and Non-occluded Conditions

Traditional robot arm grasping systems commonly rely on geometry-only decision strategies based on visual perception. Such methods have the advantage of fast response and low computational cost, which makes them suitable for deployment on edge devices. However, their performance is highly dependent on the quality of

instance segmentation. When the target object is partially occluded, the segmentation mask often becomes fragmented or incomplete, which can directly lead to incorrect grasping decisions.

To address this limitation, this study introduces a semantic reasoning module based on a LLM on top of the geometric perception layer. When instance segmentation fails under occlusion, the LLM is able to infer the relationship between the target object and surrounding occluders by jointly considering visual context and task semantics. In this way, high-level grasping decisions can be corrected even when geometric observations are incomplete.

To better evaluate how these strategies perform under occluded conditions, the experimental scenarios are categorized according to two key factors: scene attributes and decision consistency. Scene attributes describe the physical configuration of the environment, such as the degree of occlusion, the number of objects, and the spatial relationship between the target and potential occluders. Decision consistency reflects whether the system produces stable and semantically reasonable decisions across repeated observations of the same scene.

By organizing the experiments along these two dimensions, we are able to systematically compare the behavior of geometry-only strategies and the proposed LLM-assisted method under identical perception conditions. This design highlights not only the final success or failure of grasping, but also the robustness and reliability of the decision-making process in complex occluded environments.

4.2.1 Scene Categorization and Experimental Motivation

From the perspective of scene characteristics, the experimental settings can be divided into two categories:

(1) Occluded Scene: When the projection region of other objects overlaps with the visible area of the target, the target’s appearance becomes incomplete. Under such conditions, instance segmentation results from the perception layer alone are insufficient to recover the full geometric contour of the target.

(2) Non-occluded Scene: The target object is fully visible. Although neighboring objects may exist, no true occlusion relationship is formed.

From the perspective of system behavior, the experimental scenarios can be further categorized according to whether the outputs of the traditional Geometry-only method and the proposed Geometry + LLM method are consistent:

(1) Normal Case: Both the Geometry-only method and the proposed method produce the same occlusion judgment in the given scene.

(2) Effective Case: Due to incomplete perception or ambiguous spatial relationships, the Geometry-only method fails to produce a reasonable judgment or yields an incorrect result. In contrast, the proposed method compensates for such errors through semantic reasoning and world modeling, leading to a corrected decision.

4.2.2 Experimental Results and Analysis under Occluded Scenes and Non-occluded Scenes

4.2.2.1 Occluded Scene: Normal Case (Case A)

As illustrated in Fig. 4-3, this scenario represents a normal occlusion case in which the target object (cup) is physically occluded by the spoon. The YOLO-based perceptual module successfully detects both instances simultaneously. Given these reliable perceptual results, the Geometry-only method can infer the occlusion relationship based on the overlap between the two objects' bounding boxes in the 2D image space.

In this case, the proposed method produces the identical occlusion judgment as the Geometry-only baseline: both correctly identify that the target object is occluded. This result indicates that when perceptual inputs are reliable and spatial relationships are unambiguous, the proposed method demonstrates high consistency with the Geometry-only approach in occlusion reasoning.

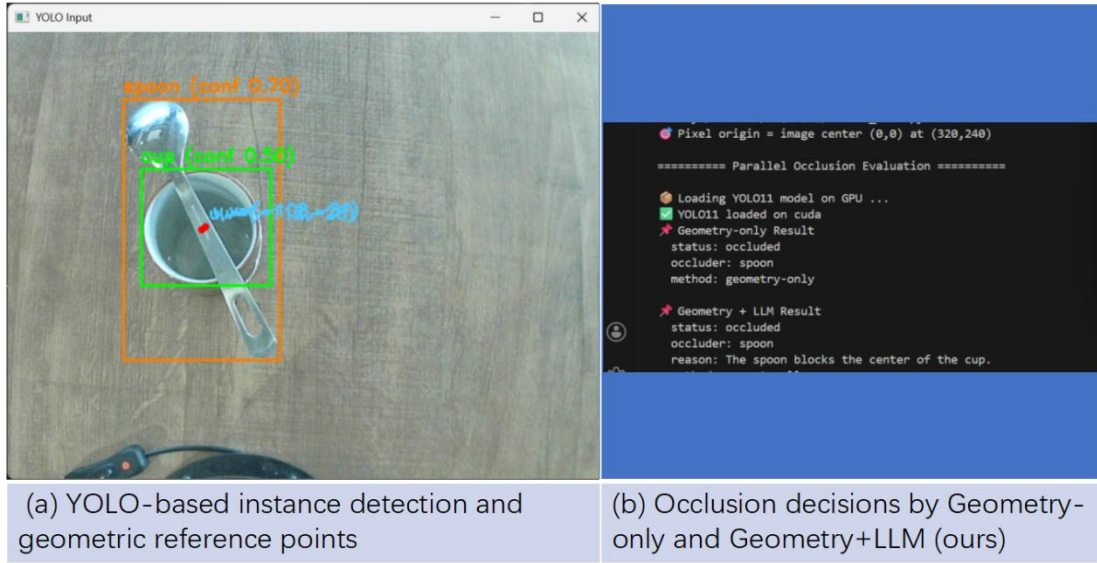


Fig. 4-3. Comparison of Perceptual Results and Occlusion Reasoning under an Occluded Scene. (a) YOLO-based instance detection results and geometric reference points, where the target object (cup) and the occluding object (spoon) overlap in the image space; (b) Comparison of occlusion judgments produced by the Geometry-only method and the proposed Geometry+LLM method under the same perceptual input.

4.2.2.2 Occluded Scene: Effective Case (Case B)

As shown in Fig. 4-4, in this scenario the spoon physically occludes the cup, causing the target to be missed by the instance detection module. As a result, only the occluding object (spoon) is detected by YOLO, and the geometric information of the target (i.e., bounding box and center point) is unavailable. Due to the absence of geometric input for the target, the Geometry-only method is unable to determine whether the target exists or is occluded. As a result, it outputs not found, and the decision process is terminated, preventing the system from proceeding with subsequent grasping strategies.

In contrast, the proposed method can still infer the occlusion relationship even when the target is not instance-detected. By leveraging the semantic attributes and spatial context of the detected spoon, together with semantic reasoning and world modeling via the LLM, the system infers that the spoon is located above the cup opening and partially occludes the target. It therefore recovers the decision that the

target exists but is occluded, and generates an appropriate subsequent handling strategy.

This scenario represents a typical Effective Case, demonstrating that the proposed method can maintain stable and coherent decision logic under incomplete perceptual conditions, without relying on the prerequisite that the target object must be fully detected.

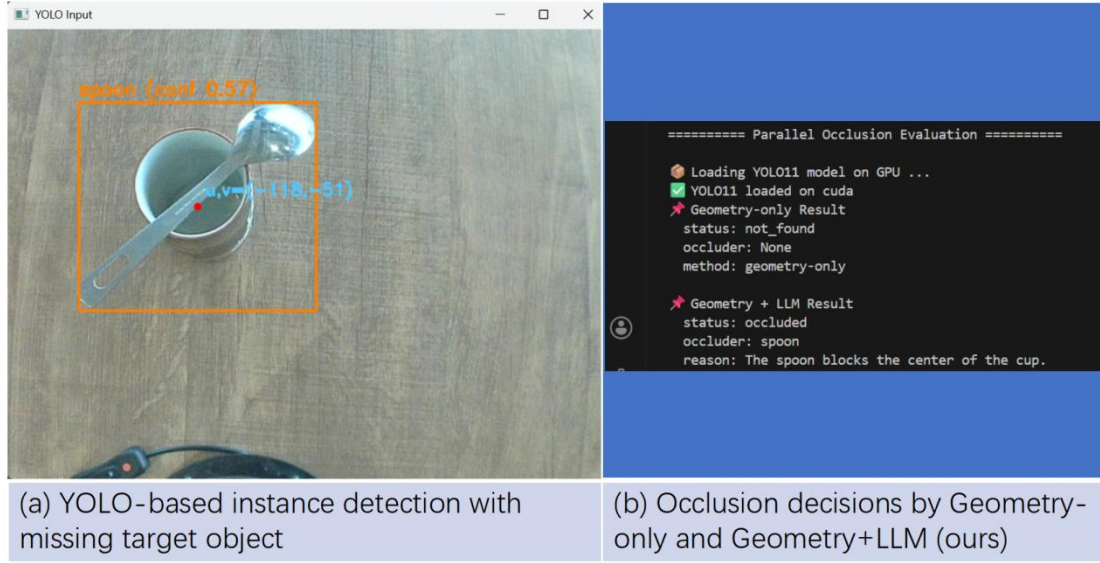


Fig. 4-4. Comparison of Perceptual Results and Occlusion Reasoning under an Occluded Scene. (a) YOLO-based instance detection results and geometric reference points, where the target object cup and the occluding object spoon overlap in the image; (b) Comparison of occlusion judgments produced by the Geometry-only method and the proposed Geometry+LLM method under the same perceptual input.

4.2.2.3 Non-occluded Scene: Normal Case (Case C)

As shown in Fig. 4-5, in this scene YOLO can simultaneously detect both the target object cup and the nearby object spoon. Although their bounding boxes appear spatially close in the image plane, the spoon does not overlap the cup region and therefore does not constitute a true occlusion.

Based on the relative positions and overlap ratio of the bounding boxes, the Geometry-only method correctly judges that the target is not occluded and outputs target (not occluded). Under the same perceptual input, the proposed method also

incorporates the semantic interpretation of spatial relationships provided by the LLM and reaches the same conclusion, namely that the target object exists and is not occluded. This result indicates that when perceptual information is reliable and spatial relationships are unambiguous, both methods yield consistent occlusion judgments.

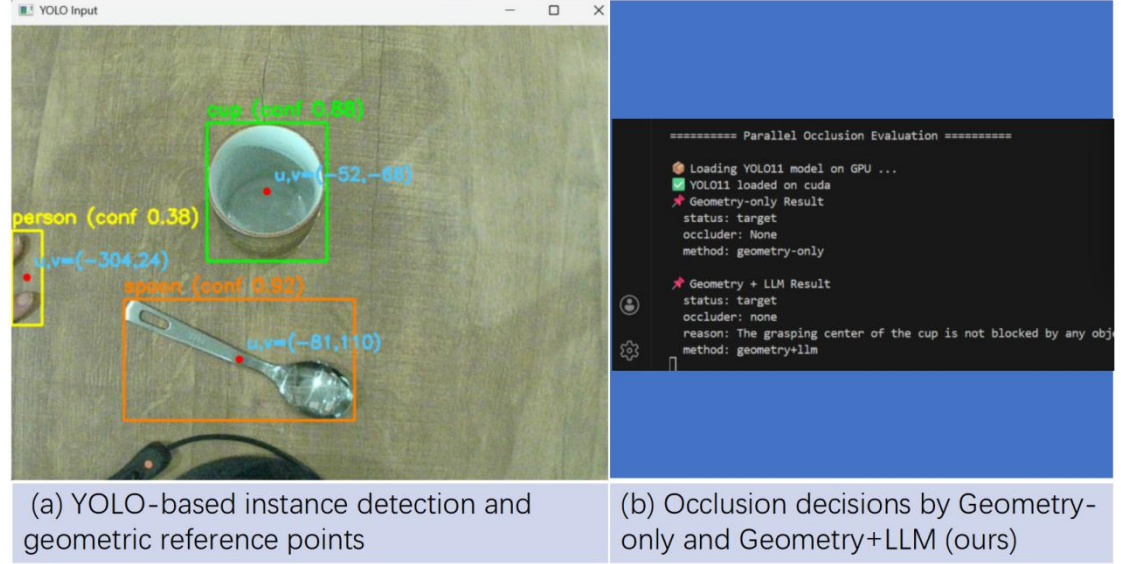


Fig. 4-5. Comparison of Perceptual Results and Occlusion Reasoning under a Non-Occluded Scene. (a) YOLO-based instance detection results of the target object cup and the nearby object spoon; (b) Occlusion judgments produced by the Geometry-only method and the proposed Geometry+LLM method under the same perceptual input.

4.2.2.4 Non-occluded Scene: Effective Case (Case D)

As shown in Fig. 4-6, in this scene the target object is fully visible, and YOLO successfully detects both the cup and the spoon. Although their bounding boxes appear spatially close and even slightly overlapping in the image plane, the spoon does not actually cover the visible region of the cup.

Under this condition, the Geometry-only method, which relies solely on geometric heuristics such as IoU or center-distance thresholds, tends to misclassify this neighboring relationship as an occlusion. In contrast, the proposed method, under the same perceptual input, compensates for this ambiguity through semantic reasoning, correctly inferring that the spoon does not block the cup and revising the decision to

not occluded.

This case demonstrates that the proposed approach is not only capable of recovering valid decisions under incomplete perception, but also produces occlusion judgments that better reflect the true spatial structure of the scene, thereby avoiding false occlusion caused by mere spatial proximity.

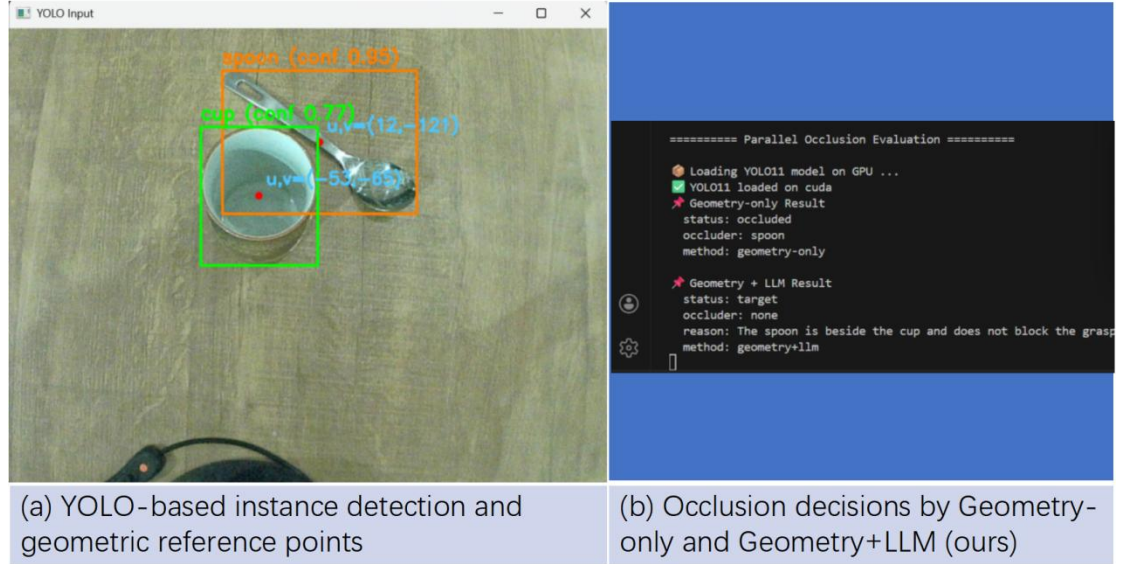


Fig. 4-6. Comparison of Perceptual Results and Occlusion Reasoning under a Non-Occluded Scene. (a) YOLO-based instance detection results of the target object cup and the nearby object spoon; (b) Occlusion judgments produced by the Geometry-only method and the proposed Geometry+LLM method under the same perceptual input.

4.2.3 Performance Summary under Occluded Conditions

Based on the above experiments, it can be observed that when perceptual information is reliable, the proposed method maintains decision behaviors consistent with traditional geometry-based approaches. In contrast, under conditions where perception is incomplete or spatial relationships are ambiguous, the system is able to compensate for and refine occlusion judgments through semantic reasoning and world modeling mechanisms. This capability endows the system with stronger decision robustness and interpretability in complex real-world environments.

4.3 Experiment II: Perceptual Ambiguity under Illumination in Occluded and Adjacent Scenarios

4.3.1 Experimental Motivation and Definition of Influencing Factors

The representative scenario-based analyses in Section 4.2 demonstrate that under conditions of incomplete perception or ambiguous spatial relationships, the proposed system can compensate for and refine occlusion judgments through semantic reasoning and world modeling. However, the stability of this strategy is highly sensitive to variations in environmental conditions.

Among these factors, illumination plays a critical role in visual perception quality. Degraded lighting conditions reduce the reliability of the perception layer, leading to lower detection confidence and shifts in bounding box locations, which in turn affect the system’s overall scene interpretation and decision-making performance.

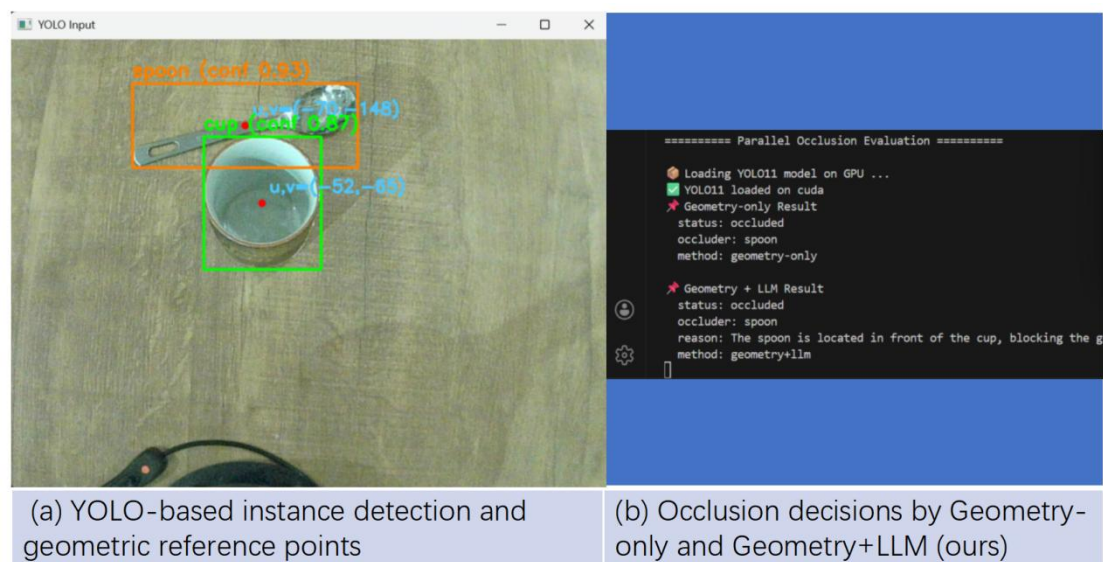


Fig. 4-7. Occlusion Misjudgment Caused by Perceptual Errors under Low-Light Conditions. (a) YOLO-based instance detection results and geometric grasp anchor points under low illumination; (b) Occlusion judgments produced by the Geometry-only and Geometry+LLM methods.

As illustrated in Fig. 4-7, reduced illumination weakens the contrast along the cup

boundary, which results in both expansion and spatial displacement of the detected bounding box. Consequently, the associated geometric grasp anchor drifts toward the region of the spoon, ultimately causing the system to falsely infer an occlusion.

4.3.2 Experimental Data and Scene Construction

To evaluate the performance differences between occlusion judgment strategies under varying illumination conditions, two representative types of scene datasets were constructed. Multiple samples were generated by systematically varying the relative spatial relationships between the target object and surrounding objects. In all experiments, the cup was designated as the target, while spoon, fork, and other tableware were used as distractors. All samples were captured using an RGB camera from multiple viewing angles under three predefined illumination ranges (bright, normal, and dim conditions). By varying both the camera viewpoint and illumination intensity, the dataset was designed to simulate realistic perceptual uncertainty and to evaluate the robustness of occlusion reasoning under changing projection geometry and degraded visual input.

4.3.2.1 Scene Types

Occluded Scene: An occluding object partially covers the target region, causing incomplete visual appearance and potentially leading to missing or degraded detection results.

Near-but-not-occluded Scene: The distractor is spatially close to the target. Although their bounding boxes may appear close or slightly overlapping in the 2D projection, the distractor does not cover the target region and therefore does not form a true occlusion.

To investigate the influence of illumination on occlusion reasoning, illumination intensity was selected as the sole control variable. By systematically varying lighting conditions to degrade visual input quality, we compare the robustness of the Geometry-only baseline and the proposed Geometry+LLM method under different

illumination levels.

4.3.2.2 Illumination Control and Calibration

A single fixed light source was used to control illumination, with an LED lamp serving as the only source of light. During experiments, indoor ceiling lights were turned off and natural light was blocked to avoid uncontrolled environmental interference. The lamp was fixed approximately 30 cm above the tabletop, while the camera was positioned slightly above and to the side of the lamp, maintaining a constant top-down viewing angle.

Illumination intensity was quantified using illuminance as the physical measurement indicator, with units of lux (lx). During measurement, a smartphone-based lux meter was placed in the vicinity of the camera. Based on the calibration results, the ambient lighting conditions were strictly categorized into three levels, as shown in Table 4-1.

Table 4-1 Classification of Ambient Illumination Levels

Illuminance Grade	Illuminance Range
Bright	>2000 lx
Normal	1000-2000 lx
Dim	0-1000 lx

4.3.2.3 Data Collection and Annotation

In this experiment, illumination intensity is selected as the only control variable. To ensure consistency across samples, all images are captured under the same tabletop layout, identical camera parameters, and a fixed viewing angle.

For each occlusion scenario, different illumination levels are generated by manually adjusting the brightness of a desk lamp. Under each combination of scene type and illumination level, 30 image samples are collected, resulting in a total of 180 images.

During data acquisition, the relative positions of the target object (cup) and

surrounding distractors (e.g., tableware) are kept strictly unchanged, ensuring that illumination is the only factor that varies across samples. All collected images are then manually annotated and treated as ground truth. Each image is labeled according to the following three aspects:

- (1) Target Existence: whether the target cup is present in the scene;
- (2) Occlusion Status: whether the target is occluded by another object;
- (3) Occluder Class: if occluded, the category of the occluding object

These annotations serve as the reference for comparing the proposed Geometry+LLM method with the traditional Geometry-only baseline under different illumination conditions. The evaluation focuses on the robustness of the semantic compensation mechanism under perceptual degradation, particularly in low-light environments.

4.3.3 Comparison Methods

This experiment is conducted on the dataset constructed in Section 4.3.1 to compare two high-level occlusion reasoning strategies. For each test image, both the traditional Geometry-only method and the proposed Geometry+LLM method are applied, and their outputs are compared with the ground-truth labels.

The Geometry-only method relies solely on geometric cues, such as the target center point and the bounding box overlap ratio (IoU) provided by YOLO, to determine whether the target exists and whether it is occluded. In contrast, the proposed method uses the same perceptual inputs but further incorporates semantic reasoning and world modeling based on a LLM to jointly infer the target existence, occlusion status, and the class of the occluding object.

To ensure that the observed behavioral differences reflect the characteristics of the high-level reasoning strategies, both methods are applied to exactly the same input images. No robotic motion execution is involved in this experiment; only the outputs of the decision layer are analyzed.

4.3.4 Definition of Evaluation Metrics

This section evaluates three metrics: target existence rate and occlusion judgment accuracy. To assess the decision performance of different occlusion reasoning strategies under incomplete perceptual conditions, quantitative evaluation metrics are designed from three perspectives: target existence judgment, occlusion relationship inference, and occluder identification.

By jointly evaluating these metrics, the robustness and decision consistency of different occlusion reasoning strategies under degraded perceptual conditions can be comprehensively assessed.

(1) Target Existence Accuracy

This metric evaluates whether the system can correctly determine the existence of the target object under complex occlusion conditions, and corresponds to the target existence decision function defined in Eqs. (3-11).

TargetExists (.)

The target existence accuracy is defined as:

$$\text{Accuracy}_{\text{exist}} = \frac{N_{\text{correct}}}{N} \quad (4-1)$$

where:

N : denote the total number of test samples;

N_{correct} : the number of samples with correct target existence judgments.

This metric reflects the system's ability to maintain consistent and reliable target existence decisions under incomplete perceptual conditions, thereby demonstrating the effectiveness of the proposed semantic completion and world modeling mechanisms.

(2) Occlusion Accuracy

Occlusion Accuracy is used to evaluate the system's ability to correctly determine whether the target object is occluded. This metric corresponds to the occlusion judgment function $Occluded(o^*)$ defined in Chapter 3 Eq. (3-14). The Occlusion Accuracy is then defined as:

$$Occlusion\ Accuracy = \frac{N_{occ}}{N} \quad (4-2)$$

where:

N_{occ} : denote the number of samples for which the system correctly classifies the occlusion state (occluded / not occluded);

N : denote the total number of test samples.

This metric reflects whether the system can distinguish true occlusion relationships from cases where objects are merely spatially close but do not actually occlude the target.

4.3.5 Experimental Results and Analysis

4.3.5.1 Partially Occluded Scenes

Table 4-2 Comparison of Occlusion Reasoning Performance under Different Illumination Conditions in Partially Occluded Scenes

Illuminance Range	Strategy	Target Existence Accuracy	Occlusion Accuracy
Bright (>2000)	Geometry-only	93.3%	93.3%
Bright (>2000)	Geometry+LLM	96.7%	86.7%
Normal (1000–2000)	Geometry-only	83.3%	76.6%
Normal (1000–2000)	Geometry+LLM	86.7%	70.0%
Dim (100–1000)	Geometry-only	66.7%	56.7%
Dim (100–1000)	Geometry+LLM	76.7%	76.7%

Table 4-2 presents a comparison between the Geometry-only and Geometry+LLM strategies under partially occluded scenes across different illumination conditions. In these scenarios, the target object is present but partially occluded by surrounding objects. The evaluation focuses on two metrics: target

existence accuracy and occlusion accuracy.

Under bright illumination, the Geometry-only strategy achieves 93.3% target existence accuracy and 93.3% occlusion accuracy. The Geometry+LLM strategy attains 96.7% target existence accuracy and 86.7% occlusion accuracy.

Under normal lighting conditions, both strategies demonstrate high target detection performance. Although the occlusion accuracy of the Geometry+LLM approach is slightly lower than that of the geometric baseline, the difference remains limited, indicating that geometric overlap reasoning is already reliable when visual information is clear and stable.

Under normal illumination, the Geometry-only baseline achieves 83.3% target existence accuracy and 76.6% occlusion accuracy. With the integration of semantic reasoning, target existence accuracy improves to 86.7%. However, occlusion accuracy decreases to 70.0%. This result suggests that under moderate illumination degradation, semantic inference introduces additional reasoning variability. While target detection remains stable, occlusion judgment becomes more sensitive to visual uncertainty.

Under dim illumination, the Geometry-only strategy declines to 66.7% target existence accuracy and 56.7% occlusion accuracy, reflecting the increased difficulty of reliable geometric inference in low-light environments. In contrast, the Geometry+LLM strategy achieves 76.7% target existence accuracy and 76.7% occlusion accuracy. The improvement in both metrics under low illumination indicates that semantic reasoning enhances robustness when geometric cues become weak or noisy.

4.3.5.2 Adjacent Non-Occluded Scenes

Table 4-3 Comparison of Occlusion Reasoning Performance under Different Illumination Conditions in Adjacent Non-Occluded Scenes

Illuminance Range	Strategy	Target Existence Accuracy	Occlusion Accuracy
Bright (>2000)	Geometry-only	76.7%	63.3%
Bright (>2000)	Geometry+LLM	90.0%	80.0%
Normal (1000–2000)	Geometry-only	60.0%	53.3%
Normal (1000–2000)	Geometry+LLM	70.0%	66.7%
Dim (100–1000)	Geometry-only	50.0%	43.3%
Dim (100–1000)	Geometry+LLM	60.0%	53.3%

Table 4-3 presents the comparison between the Geometry-only and Geometry+LLM strategies under adjacent but non-occluded scenes across different illumination conditions. In these scenarios, the target object is fully visible, while surrounding objects are spatially close but do not form true occlusion relationships. The evaluation focuses on target existence accuracy and occlusion accuracy.

Under bright illumination, the Geometry-only strategy achieves 76.7% target existence accuracy and 63.3% occlusion accuracy.

With the integration of semantic reasoning, the Geometry+LLM strategy improves performance to 90.0% in target existence detection and 80.0% in occlusion judgment. The significant improvement indicates that purely geometric overlap heuristics tend to misinterpret spatial proximity as occlusion, whereas semantic reasoning helps distinguish true occlusion from mere adjacency.

Under normal illumination, the Geometry-only baseline achieves 60.0% target existence accuracy and 53.3% occlusion accuracy. The Geometry+LLM strategy improves these values to 70.0% and 66.7%, respectively. The consistent performance gain suggests that semantic reasoning reduces false-positive occlusion judgments under moderate lighting degradation.

Under dim illumination, the Geometry-only strategy further declines to 50.0% target existence accuracy and 43.3% occlusion accuracy, reflecting increased ambiguity in visual perception. In contrast, the Geometry+LLM strategy achieves

60.0% target existence accuracy and 53.3% occlusion accuracy. Although overall performance decreases under low illumination, semantic reasoning maintains a consistent advantage over geometric-only inference.

4.3.5.3 Overall Observation and Methodological Difference

Across all illumination conditions, the Geometry+LLM framework consistently outperforms the Geometry-only baseline in adjacent non-occluded scenes. The improvement is particularly notable in occlusion accuracy, indicating that semantic reasoning effectively suppresses false occlusion detections caused by 2D spatial proximity.

It is worth noting that this behavior differs from the partially occluded scenario discussed in Section 4.3.5.1. In partially occluded scenes, the geometric baseline occasionally fails to detect certain occluding objects (e.g., small tableware such as forks or spoons) due to visual ambiguity. When such objects are not detected, the system cannot infer an occlusion relationship, which may incidentally align with the ground truth in non-occluded configurations. In these cases, the apparent correctness of the geometric baseline does not necessarily reflect accurate occlusion reasoning, but rather the absence of occluder detection.

Therefore, the relatively lower performance observed in adjacent non-occluded scenes provides a more direct evaluation of the system’s ability to distinguish true occlusion from mere spatial proximity. Under this setting, the contribution of semantic reasoning becomes more evident, as it reduces misclassification caused by geometric overlap ambiguity rather than benefiting from detection failure.

4.4 Experiment III: Simulation-to-Real Transfer in End-to-End Semantic Robotic Grasping

4.4.1 Experiment Objective

This experiment aims to verify the stability and reliability of the robotic grasping system proposed in this study under real-world execution conditions.

Rather than focusing on the performance of a specific module, this experiment evaluates the entire closed-loop control pipeline, from visual observation and task interpretation to motion generation and physical actuation. The experiment investigates whether the decisions produced by the system can be consistently and safely transferred from the computational layer to the physical robot arm, and how uncertainties in perception, planning, and actuation influence the final task outcome.

The goal is to identify the dominant sources of failure across the perception–decision–motion–execution chain, and to analyze their impact on the overall stability of the system. By examining both successful and failed executions, this experiment provides a system-level validation of the practical robustness and deployability of the proposed framework.

4.4.2 Experiment Setup and Procedure

The experiment is conducted on a tabletop scene containing two visually identical cups. One cup is filled with coffee (dark liquid), and the other is filled with transparent water. A spoon is placed on top of the water cup to introduce an occlusion/interference condition. The task command for each trial is a natural language instruction: “I need water.” The objective is to grasp the water cup while handling the interfering spoon in a safe and semantically consistent manner.

Closed-loop end-to-end procedure. Each trial follows the same system pipeline:

Observation. The robot arm returns to a predefined observation pose, and a fixed RGB camera captures the scene image.

Target selection and occlusion reasoning. The system interprets the command and performs object-level decision-making, including (i) selecting the correct target cup (water vs. coffee) and (ii) inferring whether an interfering object (spoon) forms a valid occlusion or obstruction.

Simulation-gated execution planning. Based on the inferred scene interpretation,

the system generates an execution strategy. In the occluded condition, the intended strategy is to remove the spoon first and then grasp the water cup. The planned action sequence is validated in a simulation stage to check motion feasibility and safety constraints.

Real-world execution and feedback. If the simulation stage is passed, the system converts the planned motion into joint commands and servo control signals and executes the action on the physical robot arm. The execution outcome is recorded, and the arm returns to the observation pose to start the next trial.

A total of 10 end-to-end trials are conducted under the same scene configuration. For each trial, the system records whether the occlusion-handling action (i.e., removing the spoon) is successfully executed and whether the final task goal is achieved. Failed trials are further analyzed through failure-mode attribution to evaluate the stability and interpretability of the proposed system in real-world conditions.

It should be noted that the ten end-to-end trials conducted in Experiment III are not intended to provide a statistically significant estimate of success probability. Instead, this experiment serves as a representative case study to validate the feasibility of the proposed sim-to-real transfer mechanism and to analyze system-level behavior under realistic execution conditions.

Due to the complexity of the integrated semantic reasoning, motion planning, and physical execution pipeline, the primary objective of this experiment is qualitative validation and failure-mode analysis rather than large-scale quantitative benchmarking. The selected trials are therefore analyzed in detail to illustrate both successful executions and representative failure cases, highlighting the system's decision consistency, safety handling, and current limitations.

Accordingly, the results of Experiment III should be interpreted as an in-depth behavioral analysis of the integrated system rather than a probabilistic performance evaluation.

4.4.3 System-Level End-to-End Execution Case Study

This subsection verifies the stability and interpretability of the proposed system in real-world environments through a system-level end-to-end execution case.

The experiment covers the complete control pipeline, including natural language intention understanding, visual perception, decision reasoning, occlusion handling, simulation-based planning, and real robotic execution, thereby demonstrating the system's integrated capability to bridge semantic understanding and physical grasping.

As shown in Fig. 4-8, the system detects two visually identical cups placed on the table, one containing coffee and the other containing transparent water. A spoon is positioned above the water cup, forming a partial occlusion.

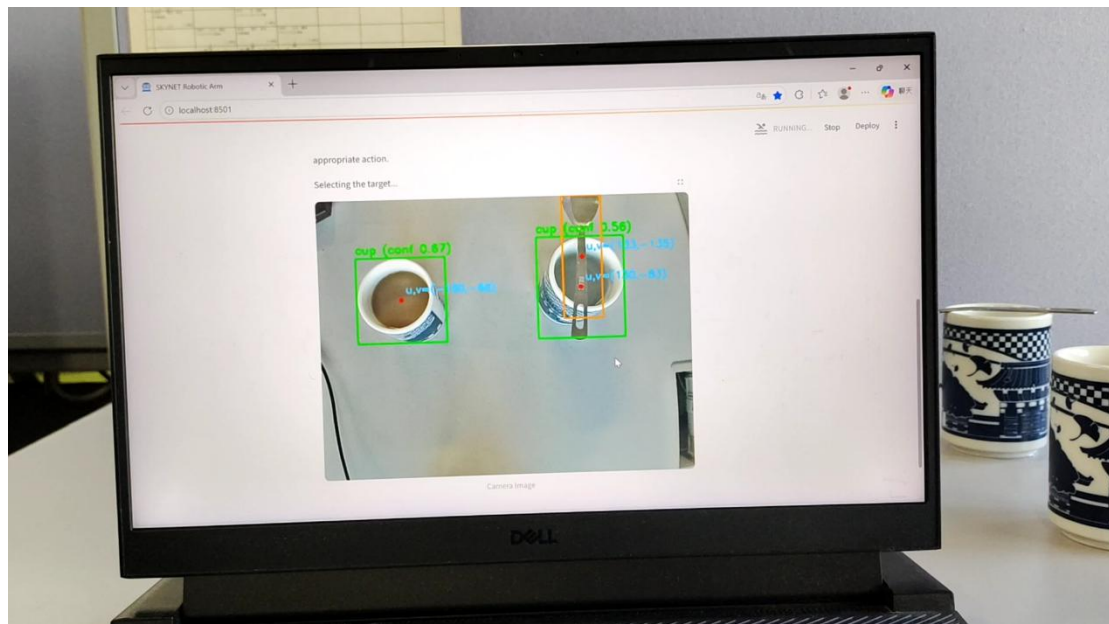


Fig. 4-8. Perception Results of the Proposed System in a Real-World

As shown in Fig. 4-9, the system first performs visual perception of the current scene and detects two visually identical cups. When relying solely on geometric information, the system is unable to distinguish their functional semantics. To resolve this ambiguity, the proposed system introduces a large language model-based semantic reasoning mechanism to interpret the visual appearance of the liquids inside the cups.

Based on this semantic inference, the system determines that the left cup contains a dark-colored liquid (coffee), while the right cup contains a transparent liquid (water). The system therefore selects the correct target—the cup on the right—and generates a corresponding control intention to rotate the target cup.

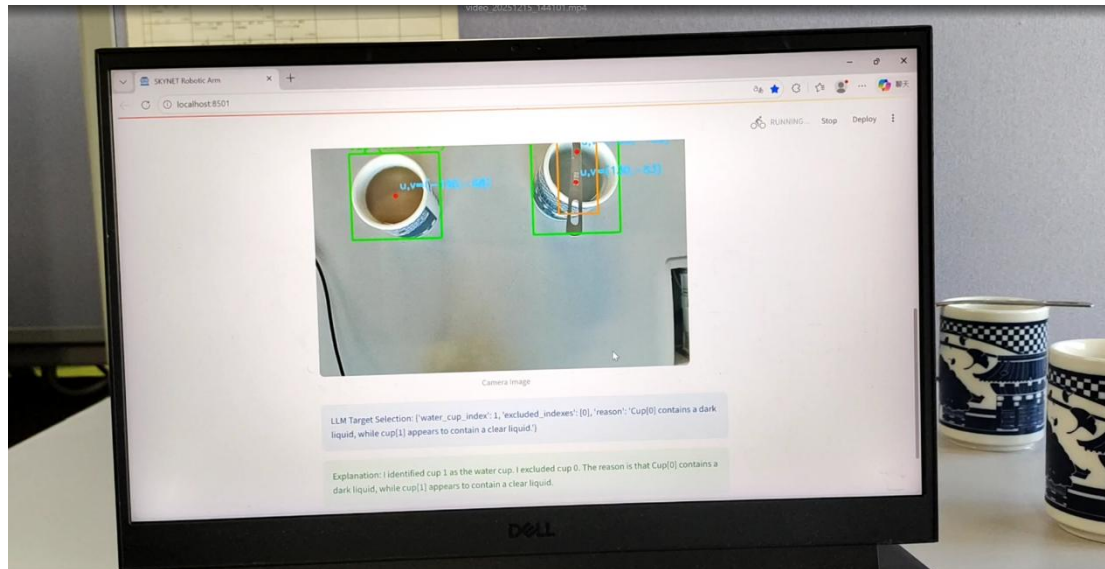


Fig. 4-9. Target Existence and Semantic Target Selection Result

As shown in Fig. 4-10, after completing target existence verification and semantic target selection, the system further analyzes the spatial relationships between the target object and surrounding objects. It determines that the target water cup is currently in an occluded state, with the spoon positioned in front of it acting as the occluding object.

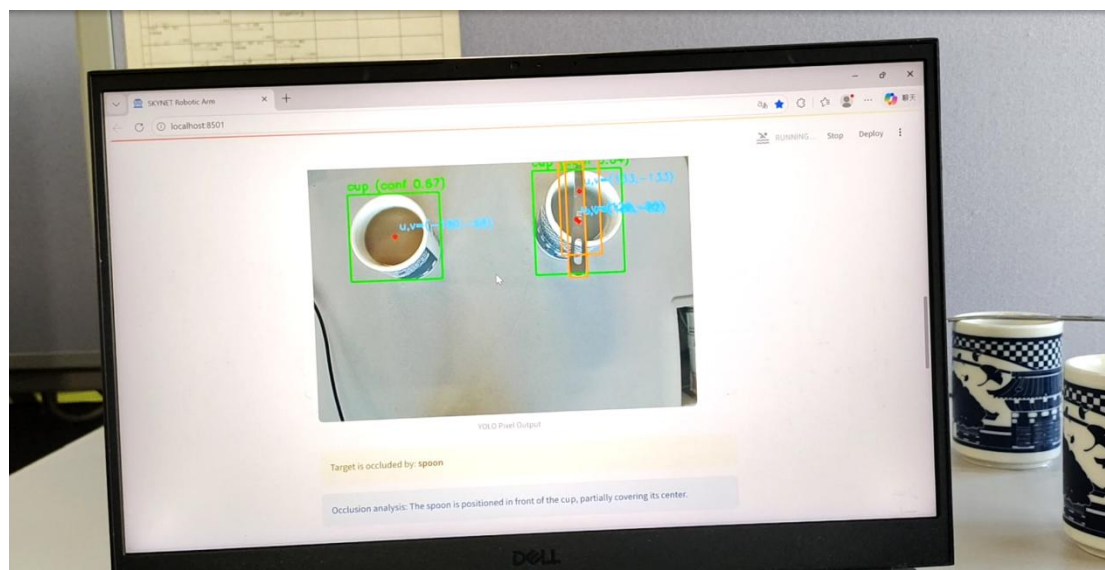


Fig. 4-10. Occlusion Judgment and Semantic Relationship Explanation

After completing the occlusion judgment, the system generates a handling strategy for the occluding object and verifies the feasibility and safety of the planned action in a simulation environment. As shown in Fig. 4-11, the system computes the corresponding joint angles of the robot arm based on the three-dimensional position of the target spoon, and plans the end-effector motion trajectory through inverse kinematics–based simulation.

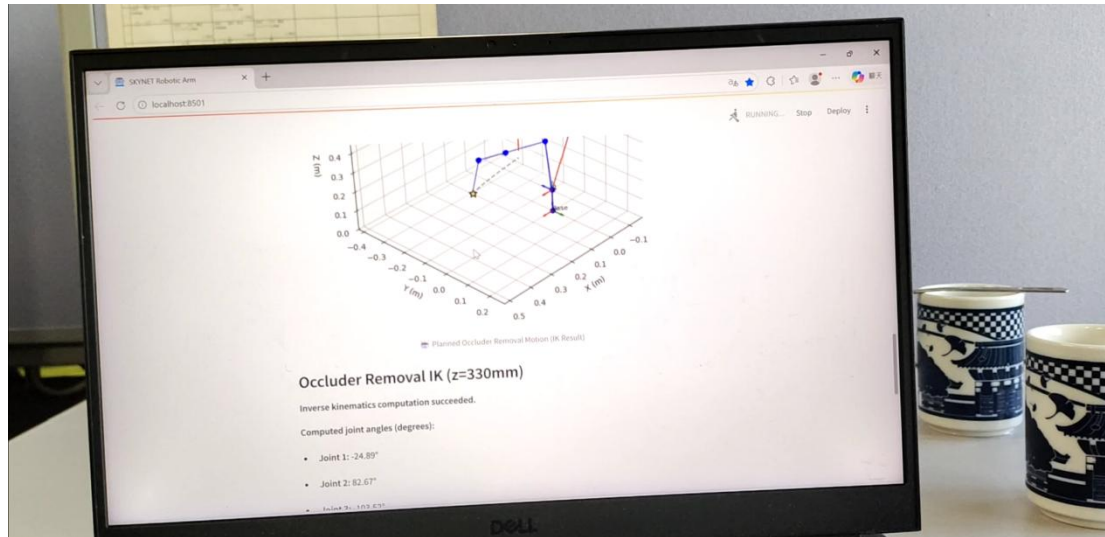


Fig. 4-11. Inverse Kinematics Simulation Result for Occluder Removal Motion

After completing the occlusion judgment and inverse kinematics–based simulation verification, the system maps the planned joint angles into servo control signals and transmits them to the physical robotic arm for execution. As shown in Fig. 4-12, the robotic arm successfully reaches the position above the spoon according to the generated inverse kinematics solution and completes the occluding-object removal action.

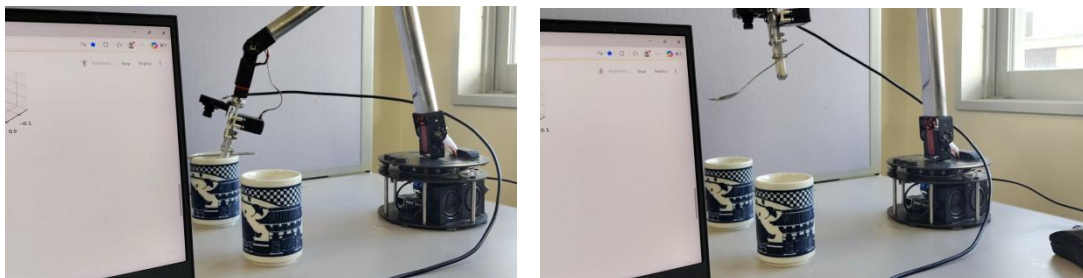


Fig. 4-12. Real-World Execution of the Occluder Removal Motion

4.4.4 System-Level Evaluation Metrics and Failure Attribution

The objective of this experiment is to verify whether, under occlusion and weak-perception conditions, the proposed system can consistently generate correct decisions and successfully execute them in the physical environment over ten trials.

In this study, a trial is regarded as an end-to-end success only if both of the following conditions are satisfied.

Correct decision-making: The system correctly selects the target object (the water cup), accurately identifies the occluding object (the spoon), and generates a reasonable occlusion-handling strategy.

Successful physical execution: The robot arm is able to execute the planned action in the real environment according to the generated plan and successfully remove the occluding object.

(1) Decision Success Rate:

This metric evaluates whether the system can correctly generate a valid high-level decision under occlusion and including target selection, before any physical execution is performed:

$$SR_{\text{dec}} = \frac{N_{\text{dec}}}{N_{\text{trial}}} \quad (4-3)$$

where:

SR_{dec} : denote the decision success rate;

N_{dec} : denote the number of trials in which the system makes a correct decision;

N_{trial} : denote the total number of experimental trials.

(2) Real-World Grasp Success Rate:

This calculates the success rate of the grasping task in the real-world environment.

$$SR_{\text{real}} = \frac{N_{\text{real}}}{N_{\text{dec}}} \quad (4-4)$$

where:

SR_{real} : denote the execution success rate;

N_{real} : denote the number of trials in which the robot physically succeeds in removing the occluder and grasping the target;

N_{dec} : denote the number of trials in which the system makes a correct decision.

(3) Decision-Conditioned Execution Gap:

$$\Delta SR = SR_{\text{dec}} - SR_{\text{real}} \quad (4-5)$$

ΔSR quantifies the execution degradation introduced by physical systems when the high-level decision is correct. It reflects the stability gap between semantic decision and physical execution

4.4.5 Experiment Data and Analysis

Table 4-4 System-Level Analysis of Decision Outcomes and Physical Grasping Result

Num	Decision Correct	Simulation Feasible	Grasp Success	Expected Outcome	Failure Cause
1	Success	Success	Failure	Failure	Machine Error
2	Success	Success	Failure	Failure	Machine Error
3	Success	Success	Failure	Failure	Machine Error
4	Failure	N.A.	N.A.	Failure	Target Selection Failure
5	Success	Success	Success	Success	N.A.
6	Success	Success	Failure	Failure	Machine Error
7	Success	Success	Failure	Failure	Machine Error
8	Success	Success	Success	Success	N.A.
9	Success	Success	Success	Success	N.A.
10	Failure	N.A.	N.A.	Failure	Target Selection Failure

Table 4-5 Simulation and Real-World Execution Performance Difference

SR_{dec}	SR_{real}	ΔSR
80%	37.5%	42.5%

From Table 4-4, we can observe the effectiveness of the proposed end-to-end system pipeline described in Section 3.5 Semantic-to-Motion Control Layer. Specifically, when the high-level decision or the simulation feasibility check fails, the system does not proceed to physical execution, which prevents unsafe or erroneous actions in the real world. This demonstrates that the system is capable of filtering out infeasible tasks before execution.

Although the decision success rate and simulation feasibility rate are relatively high, the real-world execution success rate is significantly lower. This discrepancy indicates that a correct semantic decision and a feasible simulated plan do not always guarantee successful physical execution. Most failure cases occur after a correct decision and successful simulation, but fail during real-world grasping, mainly due to mechanical system errors such as servo vibration, joint backlash, unstable motor output, and insufficient driving torque (e.g., experiments 1, 2, 3, 6, and 7).

In addition, several failures are caused by high-level semantic reasoning errors. For example, in experiments 4 and 10, the system failed to correctly identify the target and occluding objects due to visual degradation under insufficient illumination. The transparent water appeared dark in the camera image, leading the system to misclassify the scene and incorrectly conclude that no water target existed, which resulted in target selection and occlusion reasoning failure.

From Table 4-5, it can be seen that the decision success rate SR_{dec} is 80%, while the real-world execution success rate SR_{real} drops to 37.5%, resulting in a decision-conditioned execution gap ΔSR of 42.5%. This gap reflects the performance degradation introduced by physical execution and perception uncertainty, indicating that real-world mechanical constraints and visual ambiguity remain critical challenges for stable deployment.

5 Conclusions

This thesis investigates the problem of robot arm grasping decision and control under weak perception and complex occluded environments. To address the limitations of traditional control methods under incomplete observations and strong uncertainty, a hierarchical closed-loop intelligent control framework based on semantic–geometric fusion is proposed.

According to the definition of Intelligent Control presented in Chapter 1, when the controlled plant and its environment cannot be accurately modeled by conventional differential or difference equations, and when the system operates under partial observability and high uncertainty, fixed feedback or classical adaptive control strategies become insufficient. In such scenarios, the control system must integrate task planning, semantic reasoning, structural reconfiguration, learning mechanisms, and hybrid discrete–continuous control strategies, thereby enabling coordination between high-level decision processes and continuous dynamic control.

The robot arm problem addressed in this thesis exactly falls into this category of intelligent control. In real-world cluttered scenes, visual perception is often unreliable due to occlusion, illumination variation, and detection degradation, causing classical geometry-driven or rule-based controllers to frequently fail at the decision level. Therefore, this work not only involves low-level motion control and inverse-kinematics-driven physical grasp execution, but more importantly focuses on the coordination between high-level cognitive decision-making and continuous physical control under partial observability and strong uncertainty. The proposed system is essentially a belief-driven hybrid decision and control system.

To tackle this challenge, a five-layer closed-loop architecture—perception, semantic reasoning, world modeling, decision, and motion control—is constructed. Raw visual inputs are progressively transformed into structured object representations, and a grounded semantic–geometric world model is built to unify task intent, object semantics, and physical scene geometry. Based on this representation, the continuous perceptual state is abstracted into a low-dimensional discrete control state, enabling

explicit mode switching and strategy selection among different manipulation policies.

One key contribution of this work is the introduction of a hybrid control mechanism that integrates discrete decision-making with continuous motion execution. The system dynamically switches among three operational modes—abort grasping, direct target grasping, and occluder removal—according to the inferred discrete control state. This structural reconfiguration and policy switching capability embodies the core principle of intelligent control: the coordination between symbolic reasoning and continuous physical dynamics.

Furthermore, a Semantic-to-Motion Control mechanism is proposed to bridge high-level semantic decisions and low-level physical execution. By projecting image-level target points into the robot base coordinate frame through camera calibration and hand–eye calibration, and by computing feasible joint configurations via inverse kinematics and constraint validation, the framework ensures that abstract semantic decisions are consistently grounded in physically executable actions, forming a complete closed-loop control pipeline.

Under degraded perceptual conditions, such as occlusion and low illumination, the proposed framework shows clear advantages over geometry-only strategies in terms of target existence judgment, occlusion reasoning, and decision stability. Although the actual grasp success rate still has room for improvement due to mechanical errors, execution inaccuracies, and perceptual noise, the experimental analysis supports the effectiveness and robustness of the proposed hierarchical cognitive–control structure under incomplete perception.

At the same time, the experimental analysis also indicates that intelligent decision-making cannot be completely decoupled from reliable perception. Nevertheless, by introducing semantic–geometric fusion and hybrid control strategies, this work successfully elevates robot arm grasping from a purely geometric planning problem to a structured, interpretable, and extensible intelligent control problem.

In summary, this thesis demonstrates that robot arm grasping under weak perception is fundamentally an intelligent control problem that requires the coordinated integration of reasoning, world modeling, decision-making, and physical

execution. The proposed hierarchical closed-loop framework provides a systematic solution for autonomous manipulation in uncertain and partially observable environments, and lays a solid foundation for future intelligent robotic control systems.

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Appendix A

Prompt P

[System Instruction]

You are a semantic reasoning module for a robot arm system operating in cluttered and occluded environments.

Your goal is to support stable decision-making under partial observability by estimating object existence, semantic categories, and task-relevant attributes.

You must reason conservatively and provide probabilistic estimates rather than deterministic conclusions.

[Inputs]

Annotated image I': {image_with_bounding_boxes}

Observed object set O_obs:

{O_obs_list}

Task context:

{task_description}

[Task]

1. List all objects explicitly observed in the scene.
2. Infer whether additional objects may exist but are not explicitly detected due to occlusion, reflection, or limited visibility.
3. For each object (observed or inferred), estimate:
 - category $\hat{c}_j$
 - semantic attributes $\hat{a}_j$
 - semantic confidence $\hat{s}_j^{\text{sem}} \in [0,1]$

[Constraints]

Only infer objects that are visually and contextually plausible.

Do not invent objects unrelated to the scene or task.

When uncertainty is high, assign lower confidence values.

Avoid assumptions unsupported by visual evidence.

[Output Format]

Return a structured semantic world model O_{hyp} as a JSON list:

```
[  
  {  
    "id": j,  
    "category":  $\hat{c}_j$ ,  
    "attributes":  $\hat{a}_j$ ,  
    "confidence":  $\hat{s}_j^{sem}$   
  }  
]
```

Appendix B

Prompt Y

[System Instruction]

You are a Semantic Parsing Module for a robot arm system.

Your goal is to translate unstructured natural language task instructions into a structured semantic target representation.

You must focus on identifying:

The physical entity the user wants to interact with, and any functional, state, or visual constraints explicitly or implicitly mentioned.

You must reason conservatively and avoid adding unsupported assumptions.

[Input]

Task Instruction T:

{natural_language_instruction}

[Task]

Analyze the instruction T and perform the following steps:

1. Identify the target category $\hat{c}$:

Determine the general class of the object
(e.g., cup, bottle, tool, container).

2. Extract attribute constraints $\hat{a}$:

Identify all properties the object must satisfy, including:

Functional properties

(e.g., "capable of containing liquid", "used for cutting")

State / Visual properties

(e.g., "transparent", "red", "empty", "containing brown liquid")

[Constraints]

Output must be strictly structured.

Do not invent attributes not supported by the instruction.

If the instruction is ambiguous (e.g., "get that"), infer the most plausible target based on common service-robot context, but keep the description generic.

[Output Format]

Return the semantic target in the following JSON format:

```
{  
  "target_category": "cup",  
  "attributes": [  
    "capable of containing liquid",  
    "clean",  
    "accessible"  
  ]  
}
```

[Example]

Input T:

"I want to drink some water, find me a cup."

Output:

```
{  
  "target_category": "cup",  
  "attributes": [  
    "capable of containing liquid",  
    "clean",  
    "accessible"  
  ]  
}
```

Appendix C

Prompt G

[System Instruction]

You are an Occlusion Reasoning Module for a robot arm system.

Your task is to determine whether the selected target object o^* is occluded by any other object in the scene.

You must:

1. judge the occlusion relationship,
2. assign a confidence score,
- 3 provide a natural language explanation describing the spatial cause.

You must reason conservatively and rely only on visual evidence.

[Inputs]

Target object o^* :

{target_object_description}

Unified object set O_{occ} :

{list_of_candidate_objects_with_bboxes}

Annotated image I' :

{image_with_bounding_boxes}

[Task]

For each object o_k in O_{occ} (excluding o^*), determine whether it forms a valid occlusion of the target o^* .

For each candidate o_k , estimate:

- $occ_k \in \{0,1\}$: 1 if o_k visually blocks or covers a critical part of o^* ,
0 otherwise.

$\rho_k \in [0,1]$:confidence of this occlusion judgment.

e_k : a short natural language explanation describing the spatial relationship
(e.g., " o_k is positioned above the rim of o^* and blocks the opening").

[Decision Threshold]

Use the confidence threshold:

$\tau_{vlm} = 0.7$

if $occ_k == 1$ and $\rho_k \geq \tau_{vlm}$:

$valid_occlusion = True$

else:

$valid_occlusion = False$

[Constraints]

Only consider objects visible in the image.

Do not invent occluders.

If uncertain, assign low confidence.

Use spatial and visual evidence only.

[Output Format]

Return the semantic target in the following JSON format:

```
{
  "R": [
    {
      "o_star": "o*",
      "o_k": "o_k",
      "occ_k": occ_k,
      "rho_k": rho_k,
      "p_k": [u_k, v_k],
```

```

    "e_k": e_k
  }
  // one entry for each o_k ∈ O_occ \ {o*}
],
"Occluded(o_star)": Occluded(o*)
}

```

[Output Format Example]

```

{
  "R": [
    {
      "o_star": "cup_1",
      "o_k": "spoon_3",
      "occ_k": 1,
      "rho_k": 0.92,
      "p_k": [314, 227],
      "e_k": "The spoon is positioned above the rim of the cup and blocks the
opening."
    },
  ],
  "Occluded(o_star)": 1}

```

Reproducibility Note

Note that the prompt described above serves as an abstract and illustrative representation. The actual prompt used in the implemented system is more complex and may include additional task-specific constraints, structured instructions,

and iterative refinements during system development. Therefore, the prompt formulation may vary across different experimental settings.

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